Trend-Following in Financial Markets

From the P&L identity to a working system

Abstract

A trend-following system is any rule that turns past prices into a position and holds more of an asset after larger recent returns. Such a rule earns something different from what most descriptions say it earns.

The size of a trend depends on the filter chosen, so a trend is not a measurable property of a market. Two exact results describe what a trend rule earns. First, the profit of the rule "hold the asset while its price is above its moving average" is proportional to the amount by which the variance ratio exceeds one. The trading rule and the statistical test for trending measure the same quantity. So a researcher can measure the opportunity without a backtest. Second, the realised profit equals the accumulated smooth movement minus the accumulated roughness. A trend follower is long the smooth part of a price path and short the choppy part.

The edge per market is very small. A respectable diversified program corresponds to a daily return autocorrelation of about 0.0025. A decade of data from one market cannot distinguish that value from zero. The return comes from applying that edge across dozens of weakly correlated markets. The option-like payoff, with many small losses and rare large gains, is real. The rule pays for it continuously through small losses in choppy markets. The payoff fails when the rule cannot trade, as in gaps, limit moves and weekend jumps. The win rate carries no information. On a random walk, one common rule wins 41% of years and another wins 50%. Both have zero expected return. Markets drift upward and trend systems are usually long. So a backtest can profit with no predictability at all. Subtracting each market's mean return before testing removes this bias.

Volatility scaling matters more than the indicator. In one simulation it raised the Sharpe ratio from 0.04 to 0.36. Doubling or halving the look-back changed results by about 5%. Six of 10 well-known indicators are the same weighted average of past returns. Costs determine the speed of the rule. At five basis points per round trip, a 20-day rule loses money, while a 320-day rule keeps 84% of its gross return. Returns to fast trend have declined, and slow trend probably survives at a lower level. Ten years of data cannot separate a modest edge from none, so the chapter recommends deciding in advance what evidence would justify stopping.

What this is. Trend-following is the oldest systematic trading strategy still in industrial use. Writers state its theory badly more often than that of any other strategy. This chapter builds the theory from first principles. It defines the object mathematically. It identifies the quantity a trend follower earns, and shows how to estimate that quantity in practice. It then covers when and why the quantity disappears. It also covers the purpose of the surrounding risk controls.

How to read this chapter. Sections 1–3 are conceptual and historical. They cover the mental model, the candidate mechanisms, and how the industry reached its current form. Section 4 is the bibliography. Sections 5–7 contain the technical material. §5 derives what trend-following earns and what shape its payoff has. §6 surveys every serious way to estimate a trend. §7 shows that most of §6 is one estimator with three parameters. Sections 8–11 cover engineering: portfolio construction, degradation, evaluation and risk. Section 12 is the synthesis.

Different readers can start in different places.

- Readers who want only the most important idea should read §5.4, §5.5 and §7.2.
- Readers implementing a system should read §8 and §11, and treat the rest as reference.

Objectives. After this chapter, you should be able to:

- define a trend-following system as a map from price history to position, and separate trend from momentum and drift.
- derive a trend rule's expected P&L from the autocovariances of returns, and show that the canonical rule is long the variance ratio.
- decompose realised trend P&L into convexity, trend energy and realised variance, and say what each term costs.
- set a lookback from the persistence of trends rather than from a backtest, and explain why the optimum is flat.
- build the sizing and portfolio layers, including volatility scaling, the diversification multiplier and a no-trade buffer.
- test a trend system honestly, with demeaned returns, synthetic paths and a null bootstrap.

Relationship to the momentum note. A companion chapter, Momentum in Financial Markets, treats *momentum*, the statistical phenomenon of return predictability from past returns. This chapter treats *trend-following*, the trading system. The two differ, and §1.4 makes the distinction precise. The momentum chapter goes deeper on signal measurement and cross-sectional

design. This one goes deeper on the P&L functional, position sizing and risk. Each stands alone. The overlap is deliberate and small.

Epistemic tags. The chapters in this collection flag claims by status:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement.
- **[Contested]** — documented, but researchers still disagree about magnitude, robustness or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. The evidence may be private or absent. A [Practice] claim has not been debunked.

Tags appear only where the status changes what a reader should do. A tag governs the sentence or clause it opens. Untagged sentences are definitions, derivations or arithmetic. They are true by construction, not by evidence. The author checked several of the derivations numerically for this chapter. Where a number in the text comes from a simulation run for this chapter rather than from a paper, the text says so.

Notation. The table lists every symbol that recurs in the chapter, with the section that defines or first uses it. Symbols used in only one section are defined where they appear. A *bar* is the sampling interval in use.

Symbol	Meaning	Defined in
P_t; p_t	Price at time t; log price, $p_t = \ln P_t$	§1.2
r_t	One-bar log return, $r_t = p_t - p_{t-1}$	§1.2
A	Number of bars per year (252 or 256 for daily data)	§8.5
$\mathcal{F}_t$	Information set available at t	§1.2
π_t	**Position**: the signed exposure held over $(t, t+1]$, chosen using $\mathcal{F}_t$ only; in units of capital in §5	§1.2, §5.1
s_t	Raw **signal**	§5.1
x_t	Exponentially weighted trend estimate, $x_t = (1-\alpha)x_{t-1} + \alpha r_t$	§1.5, §5.5
α	EWMA decay, $\alpha \in (0,1)$, whose **span** is $2/\alpha - 1$ bars	§5.5, §6.2
w_k; $\{w_k\}$	Weight a linear filter places on r_{t-k}; the collection of weights, the **kernel**	§1.3, §5.1
$g(\cdot)$; z_t	**Response function** mapping a normalised signal to a position; the normalised signal	§1.3, §8.4
μ	Unconditional mean return (drift)	§5.2
γ_k; ρ_k	Lag-k autocovariance $\mathrm{Cov}(r_t, r_{t+k})$; autocorrelation γ_k/γ_0; $\gamma_0 = \sigma^2$ is the per-bar return variance	§5.2
$\mathrm{VR}(q)$	Variance ratio at horizon q	§5.4
$\hat{\sigma}_t$; $\sigma^\star$	Estimate, formed from $\mathcal{F}_t$, of per-bar return volatility; target volatility	§1.3, §8.3
$\tilde{r}_{i,t}$	Return of market i after the signal stage's normalisation	§1.3, §7.1
c_t; IDM	Portfolio-level multiplier; instrument diversification multiplier	§1.3, §8.6
L; n	Lookback in bars, in rules defined on returns; moving-average window, in rules defined on prices	§6.1, §6.2
μ_t; ϕ; τ	In the hidden-trend model: the unobserved drift; its autoregressive coefficient; its persistence timescale $\tau = 1/(1-\phi)$, in bars	§5.8
σ_ε; σ_η; θ	Noise volatility; volatility of the drift's innovations; per-bar signal-to-noise ratio $\theta = \mathrm{sd}(\mu_t)/\sigma_\varepsilon$	§5.8

Symbol	Meaning	Defined in
K; $\alpha^\star$	Steady-state Kalman gain; the optimal EWMA decay it implies	§5.8
N; $\bar{\rho}$	Number of instruments; average pairwise correlation *of strategy returns*	§5.9
S	Sharpe ratio of a strategy, annualised	§5.9

Several symbols carry a qualification:

- $\bar{\rho}$ is a correlation across markets, distinct from ρ_k, which is always an autocorrelation of one series.
- α is always an EWMA decay constant, never a significance level.
- σ is always a return volatility, except where subscripted as σ_η or σ_ε for model innovations.
- ε_t is generic observation noise on whichever series the surrounding model observes. It is the price in §1.4 and §6.6, and the return in §5.8.
- The *subscripted* τ_t of §1.4 and §6.7 is the smooth trend *component* of a decomposition, a different object from the unsubscripted persistence timescale τ.

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1. What trend-following is

1.1 The wrong intuition

Practitioners usually describe trend-following as some version of *"identify that a market is trending, and ride the trend."* The description is unusable rather than wrong. It assumes that "is trending" is a property a market has, and that someone could in principle detect it.

A market has no such property. An observer sees only a finite sequence of prices. Every such sequence, without exception, splits into a smooth part and a rough part in infinitely many ways. A 50-day moving average gives one trend. A 5-day moving average gives six trends in the same data. A straight line gives one trend. No decomposition is more correct than the others, because "trend" is not a feature of the data. It is a feature of the *model imposed on the data*. A price path carries no trend until a model defines one.

The second wrong intuition is physical. It comes from the word "momentum" and holds that a market in motion tends to stay in motion. A price series has no conserved quantity, no mass and no inertia. Systems built on this intuition assume persistence. Every reversal contradicts that assumption.

The third wrong intuition is the most costly in practice. It holds that trend-following is a *prediction* problem. The strategy looks like one, because it estimates something from the past and takes a position on the future. But the estimate a trend follower forms is almost never accurate enough to count as a forecast in any ordinary sense. §5.9 makes this quantitative. The return predictability that supports the entire managed futures industry corresponds to a daily return autocorrelation of

roughly 0.0025. A single market's 10-year history cannot distinguish that value from zero. As a forecast, a trend signal is terrible. Concluding that it is worthless is incorrect.

1.2 The right starting point: a map from paths to positions

The definition is deliberately mechanical:

> **A trend-following system is a function that maps the observed price history to a position, with the property that the position is an increasing function of recent price changes.**

$$\pi_t = F(p_t, p_{t-1}, p_{t-2}, ...), \qquad \frac{\partial \pi_t}{\partial r_{t-k}} \geq 0 \text{ for all } k \geq 0$$

The definition contains no claim that a trend exists, no forecast, no latent state and no regime. It is a rule, and a rule can be evaluated without assuming anything about the data-generating process. The monotonicity condition makes the rule trend-*following* rather than mean-reverting. It requires that positions increase in past returns. The requirement that π_t use only information through t makes the rule implementable.

Two questions about this function produce the main results:

1. **What it earns.** The P&L is $\sum_t \pi_t r_{t+1}$. Because π_t is a function of past returns, the expected P&L is a functional of the autocovariance structure of the return process. §5 works this out exactly. The result is that *trend-following is long autocovariance and long squared drift*. This is the central fact of the field.
2. **The shape of the payoff.** The position π_t grows with past returns, so the strategy is long when prices have risen and short when they have fallen. Its P&L is therefore a *convex* function of the price path. §5.5 and §5.6 make this precise. The convexity is exactly, and algebraically, a long position in long-horizon variance against short-horizon variance.

1.3 The master form

Essentially every trend system in production is the same four-stage pipeline. Writing the pipeline down once prevents confusion later. The "hundreds of indicators" in the practitioner literature are choices for one or two stages, with all other stages held fixed.

$$\pi_{i,t} = \underbrace{\frac{\sigma^\star}{\widehat{\sigma}_{i,t}}}_{\text{(3) risk scaling}} \cdot \underbrace{g\Big(\underbrace{\textstyle\sum_{k\geq 0} w_k \tilde{r}_{i,t-k}}_{\text{(1) kernel}}\Big)}_{\text{(2) response}} \cdot \underbrace{c_t}_{\text{(4) aggregation}}$$

Here $\tilde{r}_{i,t}$ is the return of market i at t, possibly normalised. The term c_t is a multiplier at the portfolio level. The table describes the four stages.

Stage	What it does	What varies between systems
(1) Kernel $\{w_k\}$	Summarises the recent past into one number	Shape and timescale of the weights on past returns
(2) Response $g(\cdot)$	Converts the signal into a desired exposure	Linear, capped, sign, or bump-shaped
(3) Risk scaling $\sigma^\star/\widehat{\sigma}_{i,t}$	Converts exposure into a position of known risk	Volatility estimator and target
(4) Aggregation c_t	Combines markets into a portfolio at a total risk level	Correlation handling, capital constraint, drawdown control

A 200-day moving-average crossover, a Donchian channel breakout, the sign of 12-month time-series momentum, and a Kalman-filtered local-linear-trend model are all instances of this form. §6 enumerates them. §7 shows their equivalences. Several of these equivalences are exact identities rather than analogies.

The stage that matters most is not the stage practitioners argue about. §7.3 and §8.1 argue that **stage (3) dominates**, stage (4) comes second, and stages (1) and (2) matter least. Those two stages contain the "indicator", the part of the system that has a name and attracts the most discussion. In a simulation with realistic volatility clustering, moving from a raw signal to

a volatility-scaled one took the Sharpe ratio from 0.04 to 0.36. Changing the kernel's timescale by a factor of two from its optimum cost about 5% of the Sharpe ratio.

1.4 What trend-following is not

Confusing the concepts in the table causes real losses. The middle column gives the formal definition of each object. The right column says why people confuse it with trend-following.

Concept	Formal object	Why it is confused with trend-following
Trend-following	A map $\pi_t = F(\mathcal{F}_t)$ increasing in past returns	—
Momentum	The statistical claim $\mathbb{E}[r_{t+1} \mid r_{t-L:t}]$ is increasing in past returns	Trend rules *monetise* momentum, but they also monetise drift and convexity. A profitable trend backtest is not evidence of momentum
Drift	The unconditional mean $\mu = \mathbb{E}[r_t]$	A long-only asset with positive drift produces a positive trend backtest with zero predictability (§5.3)
Trend, as a decomposition	The low-frequency component τ_t in a trend-plus-cycle-plus-noise split $p_t = \tau_t + \psi_t + \varepsilon_t$	This is a modelling choice, not an observable. A perfectly trending line plus noise has *negative* return autocorrelation
Technical analysis	A heterogeneous body of chart-pattern heuristics	Trend rules originated inside it and share its vocabulary. But a trend rule is a measurable function with a testable P&L, and a head-and-shoulders pattern is not
Market timing	Switching between an asset and cash on a signal	A long-only trend rule *is* market timing, but trend-following in general is long/short and multi-market
Managed futures / CTA	An industry and a fee structure	Most CTAs are trend-followers, so index returns are used as a proxy for the strategy. The proxy includes fees, survivorship and strategy drift
Buy-the-dip / mean reversion	$\partial \pi_t / \partial r_{t-k} < 0$	Literally the same functional with the opposite sign. The correct sign depends entirely on the horizon (§9.2)

Two of these distinctions need expansion, because they corrupt research.

Trend is not momentum, and a backtest cannot tell them apart

Take a price process that is a deterministic upward line plus independent noise:

$$p_t = \mu t + \varepsilon_t, \qquad \varepsilon_t \sim \text{iid}(0, \sigma_\varepsilon^2)$$

In the decomposition sense, this series has a *perfect* trend, because the low-frequency component is exactly μt. Its returns are $r_t = \mu + \varepsilon_t - \varepsilon_{t-1}$, an MA(1) with autocorrelation $\rho_1 = -1/2$ and $\rho_k = 0$ for $k \geq 2$. So the series has a perfect trend and *anti*-momentum: an up move predicts a down move.

Meanwhile, $r_t = \phi r_{t-1} + u_t$ with $\phi > 0$ and zero mean has momentum and no trend at all. The level has no low-frequency component.

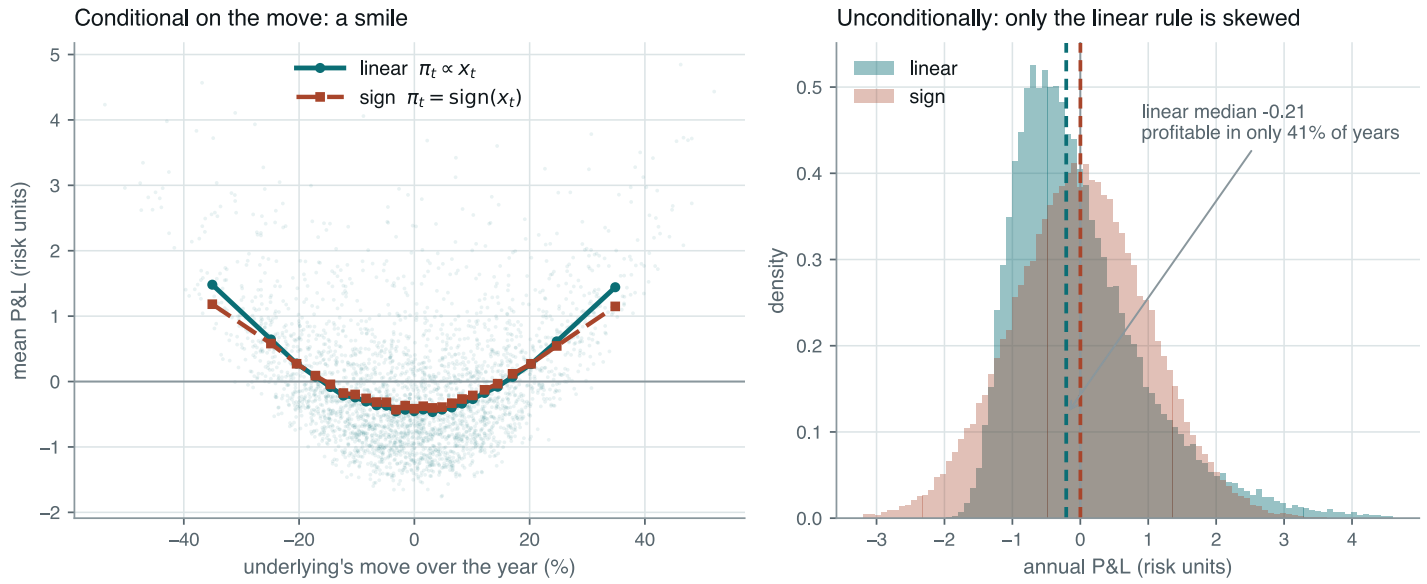
Trend is a property of the level. Momentum is a property of the returns. A trend-following *rule* does not distinguish them. It makes money from drift, from trend and from return autocorrelation, and its P&L does not reveal which. This is acceptable for earning money and harmful for research, because the three sources have very different stability properties. Drift is stable and carries little information. Autocorrelation is fragile and carries a lot. A rule that earns its profit from drift, without the researcher knowing, fails as soon as it trades a market without drift.

The convexity is not a free option

Trend-following's payoff resembles a long straddle (§5.6). Writers often present this as good news: "trend-following gives you convexity for free". The claim is wrong. Convexity is never free. Only the form of the payment varies. A straddle buyer pays a premium up front. A trend follower pays continuously, through the whipsaw losses incurred when a signal reverses. §5.5 shows this payment as a $-\sum_t r_t^2$ term in an exact algebraic identity. The strategy is profitable exactly when the profit it collects from the persistence of trends exceeds the realised variance it pays.

1.5 The payoff in one figure

Run a trend rule on paths with *no* predictability whatsoever, pure random walks, and plot what it earns. This shows most clearly what kind of object a trend rule is. The figure shows the result.



The figure comes from a simulation, run for this chapter, of 60,000 independent random-walk years with a 65-bar-span EWMA rule. Three points follow from it:

1. **Conditional on the size of the year's move, the payoff is U-shaped.** Big moves in either direction pay, and small moves cost. This is the straddle analogy, and it holds for both response functions.
2. **The mean is zero, as it must be.** The underlying has no predictability. The simulated mean P&L was $+0.003$ risk units, against a standard deviation of 1.0. None of this is edge.
3. **The linear rule's P&L is right-skewed, with a negative median.** It was profitable in only 41% of years, with a median of -0.21 risk units and a skew of $+1.40$. The sign-response rule discards the magnitude of the signal. It was profitable in exactly 50% of years, with a skew of -0.01. The symmetry follows from the structure of the rule. The sign rule's P&L is $\sum_t \mathrm{sign}(x_t) r_{t+1}$, a sum of driftless increments, each multiplied by ± 1. In continuous time, the P&L is $\int_0^T \mathrm{sign}(x_t)\, dW_t$, whose quadratic variation is exactly T. By Lévy's characterisation, that integral *is* a Brownian motion. So the sign rule's annual P&L is exactly Gaussian, with median zero, skew zero, and a win half the time.

Readers most often misread point 3. **A low hit rate is not evidence of a bad strategy, and a high hit rate is not evidence of a good one.** With zero edge, the choice of response function alone can produce either. Potters and Bouchaud (2006) derived this analytically. For a trend rule on a driftless random walk, the average gain per trade is exactly zero, while the fraction of winning trades falls below one half. With no drift, the win rate does not depend on scale, because volatility cancels out of the problem entirely. At a fixed drift, the win rate falls as volatility rises, because it depends on the ratio of drift to volatility. When diagnosing a trend system, the hit rate indicates the shape of g. It does not indicate whether the system works.

§1 Key takeaways

1. A market does not "have" a trend. A trend is a property of the imposed model. "Detecting a trend" is therefore always a modelling choice with a Type I / Type II trade-off. It is never a measurement.
2. Define the object mechanically: **a trend-following system is a map from price history to position, increasing in recent returns**. Everything else derives from that map.
3. The system is not a forecaster. The predictability it exploits is far too small to detect in one market (§5.9). It is visible only in aggregate.
4. Every production system is the same four-stage pipeline: kernel, response, risk scaling and aggregation. Named indicators are choices of the first two stages.
5. **Risk scaling matters more than the indicator.** In simulation, adding volatility normalisation moved the Sharpe ratio from 0.04 to 0.36. Halving or doubling the kernel timescale cost about 5%.
6. Trend $\neq$ momentum $\neq$ drift. A profitable trend backtest is consistent with zero return predictability, so it is not evidence of momentum.

7. The payoff is option-like, and the strategy *pays for* the option continuously, in realised variance. Convexity carries a continuing cost.
8. The hit rate is a property of the response function, not of the edge. On a pure random walk, a linear trend rule wins 41% of years and a sign rule wins 50%, with identical expected returns of zero.

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2. Why trends could exist

2.1 The objection that has to be answered first

If prices reflected all available information, past prices would be useless for predicting future ones. Every rule of the form $\pi_t = F(\mathcal{F}_t)$ would then have zero expected P&L. Trend-following uses only the price series itself, the most public dataset in existence. So it is the weakest possible violation of market efficiency. The efficient-markets hypothesis most obviously rules out this strategy.

Three replies follow, in increasing order of force.

The joint-hypothesis reply. Any test of efficiency is a joint test of efficiency and of an asset-pricing model. If trend-following earns a positive return that compensates for a risk the model does not contain, efficiency survives. This reply is logically valid and has almost no empirical content, because analysts can always postulate a risk factor. It does show that "trend-following works" and "markets are efficient" are not formally contradictory.

The Grossman–Stiglitz reply. If prices were perfectly informative, no one would pay to gather information, and so prices could not be informative. Equilibrium therefore requires prices to be *slightly* wrong. The error must be large enough to cover the marginal information-gatherer's costs, and no larger. The equilibrium level of inefficiency is positive. At that level, arbitrage is barely worth doing after costs. Trend-following's measured edge, a per-market Sharpe ratio of order 0.2–0.4 before costs (§5.9), is roughly the size that "barely worth doing" implies.

The empirical reply. Researchers have documented the effect over very long samples, including in markets that did not exist when the rules were designed. Hurst, Ooi and Pedersen (2017) report positive trend-following returns across 67 markets back to 1880. Lempérière and co-authors (2014) remove the markets' own drift from spot commodity and index series. They find a t-statistic of about 5 since 1960, and about 10 since 1800. **[Contested]** Both are practitioner studies by firms that sell the strategy, which is a real reason to discount them. These firms are also the only groups with the data, and no one has overturned their long-horizon results. Against them, Huang, Li, Wang and Zhou (2020) argue that drift confounds the canonical time-series momentum test, the same drift those papers try to remove. After correcting for it, they find little evidence of the effect (§5.3). Researchers have not resolved the disagreement. It concerns specification, not data.

None of this explains *why* trends exist. Researchers have proposed five families of mechanism, and the subsections below take them in turn. The families are not mutually exclusive. This chapter takes the view that all five contribute, in proportions that vary by asset class and horizon.

2.2 Family A — slow information diffusion

[Hypothesis, well-formalised] Information reaches market participants gradually rather than instantaneously. If news reaches one group at t and another at $t + k$, the price adjusts in steps. A sequence of partial adjustments in the same direction is precisely positive return autocorrelation.

Hong and Stein (1999) formalise this with two types of agent. "Newswatchers" trade on private information but ignore prices. "Momentum traders" trade on price changes but ignore fundamentals. Slow diffusion among newswatchers produces underreaction, which momentum traders then amplify into overshooting and eventual reversal. The model is attractive because a single assumption generates the whole empirically observed sign pattern (§9.2), not just one piece of it. The pattern is underreaction at short horizons, momentum at medium horizons and reversal at long horizons.

What would falsify it: trends should be stronger where information diffuses more slowly, in smaller, less-covered and less-liquid markets. Equities broadly support this: momentum is stronger in small caps and in names with low analyst coverage. As an explanation for trends in Treasury futures or major FX, diffusion is much weaker. These are the most information-efficient markets in existence, and trend-following nonetheless works well in them. This gap is the strongest evidence that diffusion is not the whole explanation.

2.3 Family B — behavioural biases

[Hypothesis] Several biases documented at the individual level produce slow price adjustment in aggregate:

- **Anchoring and conservatism.** Barberis, Shleifer and Vishny (1998) model investors who update too slowly from a prior. That produces underreaction to individual news and overreaction to long streaks.
- **Overconfidence and biased self-attribution.** Daniel, Hirshleifer and Subrahmanyam (1998) model investors who overweight private signals and attribute confirming public news to their own skill. That produces continued overreaction and eventual correction.
- **The disposition effect.** Investors sell winners too early and hold losers too long. Grinblatt and Han (2005) show that this creates a spread between the market price and a reference price, and that the spread predicts returns. The selling pressure on winners slows the adjustment, so the price drifts toward fundamental value slowly rather than jumping.
- **Extrapolative expectations.** Greenwood and Shleifer (2014) show from survey data that investors expect *high* returns after high past returns. A rational model with mean-reverting returns predicts the opposite. Barberis and co-authors (2015) build an equilibrium model on this finding.

The problem with this family is the step from individual bias to price impact. The step requires the biased agents to be the marginal price-setters. In Treasury futures, they are not. The biases themselves are real and well documented at the individual level. Behavioural explanations account for more in equities and much less in the futures markets where trend-following actually earns most of its money.

2.4 Family C — mechanical and flow-driven persistence

[Fact for the mechanism, [Hypothesis] for its share of the effect] This family requires no irrationality at all. Several structural features of markets mechanically generate serially correlated order flow, and serially correlated order flow generates serially correlated returns.

- **Metaorder execution.** Traders cannot execute a large institutional order at once. They split it into child orders over hours or days, and each child order pushes the price in the same direction. The empirical impact of a metaorder of size Q scales as $\sqrt{Q}$. Tóth and co-authors (2011) document this "square-root law" across markets and brokers. The impact accumulates over the execution horizon. This accumulation is exactly a short-horizon trend.
- **Risk-management flows.** Volatility targeting, portfolio insurance, stop losses, margin calls and value-at-risk limits all force selling into declines and buying into rallies. Brunnermeier and Pedersen (2009) formalise the destabilising feedback. Losses tighten funding. Tighter funding forces deleveraging. Deleveraging causes losses.
- **Systematic rebalancing.** Leveraged ETFs must rebalance daily in the direction of the day's move. Target-date and risk-parity funds rebalance on schedules. These are price-taking flows with known signs.
- **Central-bank and macro persistence.** Policy rates move in cycles lasting quarters to years, and inflation and growth are persistent. Futures markets on rates, bonds and currencies inherit that persistence directly. This is the most plausible single explanation for why trend-following works best in fixed income and FX. It does not involve market inefficiency at all.

[Hypothesis] This chapter takes the view that this family is the most likely primary driver in futures markets. The family predicts that trend-following works wherever large slow flows exist, regardless of investor sophistication. The cross-section of results matches this prediction.

2.5 Family D — risk transfer and hedging pressure

[Contested] Futures markets contain natural hedgers with inelastic demand, such as a producer who must sell forward or an airline that must buy fuel. Keynes argued that the speculators who take the other side need compensation. That produces a risk premium tied to the direction of hedging pressure. De Roon, Nijman and Veld (2000) find hedging-pressure effects in futures returns across markets.

This mechanism explains a *return*. It explains a trend only if hedging pressure is itself persistent. Persistence is plausible, because a producer's hedging need changes with the slow-moving fundamentals of the business. On this account, trend-following does not exploit an error. Trend followers receive payment for holding risk that someone else must shed, and the payment lasts as long as the need does.

This account has an implication for *convexity*. A liquidity provider to forced sellers earns a premium and takes tail risk. A trend follower buys convexity and pays a premium. The exposures are opposite, so hedging pressure cannot be the whole explanation. It may still account for a large part of the *level* of returns, in commodities specifically.

2.6 Family E — there is no effect, only drift and selection

[Contested] The nihilist position deserves its strongest statement, because no obvious error refutes it.

Most assets have positive expected returns. A trend rule applied to an asset with positive drift will be long most of the time, so it will earn the drift. §5.3 shows that this effect makes the standard time-series momentum test *positively biased*: $\mathbb{E}[r_{t+1} \cdot \text{sign}(r_{t-L:t})] > 0$ whenever $\mu > 0$, even under complete independence. Huang and co-authors (2020) make this argument formally. In their sample, the time-series momentum effect largely disappears once the analysis controls for the unconditional mean.

A second problem is selection. Decades of practitioner search selected the strategy's parameters over the same price histories now used to validate it. This is the definition of data snooping. Sullivan, Timmermann and White (1999) applied White's Reality Check to a universe of nearly 8,000 technical trading rules. Rules that looked strongly significant in isolation were not significant once the test accounted for the search. Brock, Lakonishok and LeBaron (1992) had earlier found technical rules profitable on the Dow over 1897–1986. The snooping critique targets results of that kind. The survey of the whole technical-analysis literature by Park and Irwin (2007) concludes that data snooping and ignored transaction costs substantially compromised the early positive results.

A third problem is delivery. CTA returns that investors actually received are far worse than the strategy's paper returns. Bhardwaj, Gorton and Rouwenhorst (2014) find that investors in commodity trading advisors earned close to nothing after fees. Both the funds' gross performance and its persistence were weak once the authors removed survivorship and backfill biases.

The view taken here. Families C and D are probably real and probably small. Family E is the correct null. The long-history studies have weakened it but not defeated it. Families A and B do genuine work in equities and less elsewhere. The claim that this evidence defends is narrow. *Futures markets exhibit weak positive return autocorrelation at horizons of weeks to months. The autocorrelation is large enough to support a diversified gross Sharpe ratio of roughly 0.7–1.0. Investors' experience net of fees is much worse than that.* Anyone who claims more should show evidence that survives the drift adjustment.

2.7 Telling the families apart

Claims about mechanisms have value only if they make differing predictions. These families do, as the table shows.

Family	Predicts trend is stronger…	Predicts trend is weaker…	Distinguishing test
A. Slow diffusion	in less-covered, less-liquid, more-opaque markets	in the most efficient markets	Cross-sectional: sort markets by information efficiency
B. Behavioural	where retail and unsophisticated flow dominates	in institutional-only markets	Sort by investor composition; test disposition-effect proxies directly
C. Mechanical flow	where large slow flows exist, regardless of sophistication	where flows are small or fast	Use positioning and metaorder data; test whether trend returns concentrate around known flow events
D. Hedging pressure	in physical commodities with concentrated hedgers	in financial futures	Commitments-of-Traders positioning as a conditioning variable
E. Drift plus snooping	nowhere — the effect is an artefact	everywhere, after correction	Demean returns before testing; out-of-sample on markets and eras not used in rule design

The cheapest test that separates the families is family E's. [Practice] **Recommendation: demean the returns market by market before running any trend test.** A surviving effect merits further study. An effect that does not survive is a rediscovery that stocks and bonds went up. The test costs one line of code, and it eliminates the most common way that trend research misleads its author.

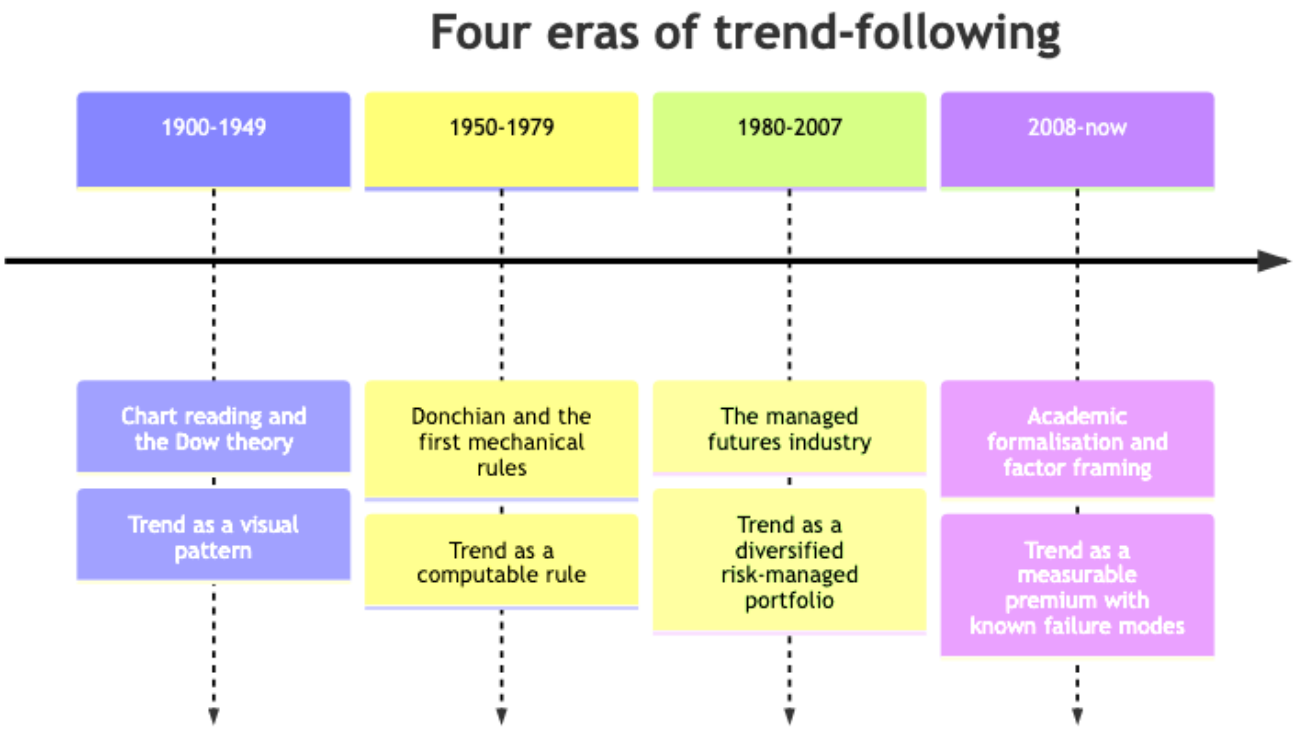
§2 Key takeaways

1. Trend-following is the weakest possible violation of efficiency, because it uses only public prices. So it requires a mechanism as well as a backtest.
2. Grossman–Stiglitz shows that equilibrium *requires* a small residual inefficiency, which is not an anomaly. The measured size of the trend effect is about the size that "barely worth arbitraging" implies.
3. Five families of mechanism exist: slow diffusion, behavioural bias, mechanical flow, hedging pressure, and drift with data snooping. They are not mutually exclusive, and each explains a different subset of the evidence.
4. Behavioural and diffusion explanations account for the most in equities and the least in Treasury and FX futures. Trend-following earns most of its money in those futures. Treat that gap as unresolved.
5. **The view taken here is that mechanical flow (family C) is the most likely primary driver in futures.** It predicts trends wherever slow flows exist, regardless of investor sophistication, which matches the cross-section.
6. The nihilist null, drift plus data snooping, has not been defeated. Long-history evidence, produced by interested parties, has weakened it.
7. **Demean each market's returns before testing.** It takes one line of code, and it removes the most common way researchers mislead themselves in this literature.

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3. How the practice evolved

The history of trend-following is unusual. The practice ran roughly 40 years ahead of the theory, and the two communities barely spoke. Systematic futures traders ran volatility-scaled multi-market trend portfolios in the 1970s. The first academic paper to describe that object as a factor appeared in 2012. The history is worth reading in order mainly because it explains why the standard implementation has its current form. Most of its features respond to past failures. The timeline shows the four eras.



3.1 Era I — Chart reading (1900–1949)

- **Contribution.** Price movements have a direction that persists, and a participant should align with it rather than oppose it. Dow's editorials introduced primary and secondary trends, and the confirmation of one index by another. Others later systematised these ideas as "Dow theory".
- **What changed.** Observers could now form a systematic opinion about trend, instead of describing it after the fact.
- **Limitations.** Nothing here was a rule. Every judgement, such as whether a trend was primary or confirmed, required a human. So nobody could test the method, and nobody could attribute its results.
- **Lasting influence.** The influence is small and consists mostly of terminology. The specific patterns, such as head and shoulders and triangles, have not survived quantitative testing. Lo, Mamaysky and Wang (2000) gave the patterns the fairest possible test, using kernel smoothing. They found weak statistical content that was mostly not tradeable.

3.2 Era II — Mechanical rules (1950–1979)

- **Contribution.** Richard Donchian's moving-average and channel-breakout rules turned trend-following into arithmetic. Anyone could compute the rule on a schedule, without judgement. Donchian's 5- and 20-day moving-average crossover and the "n-day high" breakout are still the reference implementations.
- **What changed.** Once the rule was mechanical, analysts could backtest it and attribute its P&L to the rule rather than to the trader. This is the most important methodological step in the field's history.
- **Limitations.** The rules were single-market and fixed-size, with no volatility scaling and no portfolio construction. Position size was a fixed number of contracts. So risk varied by a factor of five across markets, and across time within one market.
- **Lasting influence.** The influence is large. Every rule in §6.1–§6.5 descends from Donchian's rules, and the breakout rule in particular is essentially unchanged.

3.3 Era III — The industry (1980–2007)

- **Contribution.** Firms built trend-following into a *portfolio* discipline. The discipline used many markets, positions sized by volatility, risk budgeted at the level of the book, and systematic roll and execution. The firms, such as Campbell, Millburn, AHL, Chesapeake, Winton, Aspect and the Dennis–Eckhardt "Turtle" cohort, developed this largely in private and largely without publishing.

- **What changed.** The firms realised that the edge per market is small and that the portfolio effect supplies most of the return. A per-market Sharpe ratio of 0.3 across 50 weakly correlated markets gives a portfolio Sharpe ratio of 0.7 to 1.1, depending on how weak the correlations are (§5.9). No single market's rule needed to be good.
- **Limitations.** The firms published almost none of it, so the evidence base was track records. Track records contain survivorship, selection and opaque fees. Bhardwaj, Gorton and Rouwenhorst (2014) later showed how large those biases were.
- **Lasting influence.** The firms invented the modern implementation in this era. Volatility targeting, correlation-aware sizing, ensembles across timescales and systematic roll all date from it. §8 essentially describes what these firms worked out.

3.4 Era IV — Formalisation (2008–present)

- **Contribution.** Academic and quantitative-practitioner work made the object measurable. Moskowitz, Ooi and Pedersen (2012) introduced "time-series momentum" as a factor across 58 futures markets. Fung and Hsieh (2001) had already shown that trend followers' returns load on lookback straddles rather than on linear asset exposures. Potters and Bouchaud (2006), Bruder and Gaussel (2011) and Dao and co-authors (2016) derived the P&L decomposition that §5 reconstructs. Levine and Pedersen (2016) and Beekhuizen and Hallerbach (2017) showed that the various trend rules are one filter with different weights.
- **What changed.** Trend-following stopped being a strategy with a track record. It became a factor with a functional form, a risk decomposition, a set of known failure modes, and a price. Its fees also fell sharply at that point.
- **Limitations.** The formalisation arrived alongside a decade of poor realised performance (2009–2019). It is unclear how much of the formalisation describes a real premium, and how much gives a well-specified account of a data-snooped one (§9.4).
- **Lasting influence.** The influence is ongoing. Current work includes changepoint-aware signals, learned response functions and cost-aware dynamic portfolios. All of it builds on this era's framing.

§3 Key takeaways

1. The practice preceded the theory by roughly 40 years. Practitioners invented almost everything in a modern system before anyone described it.
2. Donchian contributed *mechanisation*, not a better indicator. Mechanisation made backtesting and attribution possible.
3. The industry's key insight was that the per-market edge is small and the portfolio effect supplies most of the Sharpe ratio. Design for the portfolio.
4. Almost all the evidence from Era III was track records. So the critiques of survivorship and fees in the 2010s undermined much of that evidence.
5. The formalisation from 2008 onward coincided with a decade of weak performance. Whether that is coincidence or an effect of the formalisation is the open question of §9.

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4. Foundational references

This section annotates the references and groups them by how a reader should use them. Every entry links to a freely readable copy where one exists.

4.1 The P&L and payoff structure

- **Fung, W. & Hsieh, D. A. (2001).** "The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers." *Review of Financial Studies* 14(2), 313–341. [DOI] — This paper established that trend followers' returns are option-like, and modelled them with lookback straddles. Every "crisis alpha" claim derives from it.
- **Potters, M. & Bouchaud, J.-P. (2006).** "Trend followers lose more often than they gain." *Wilmott Magazine.* — A short, exact paper that best resolves confusion about hit rates. On a random walk, the average gain per trade is zero while the win rate falls below one half.
- **Bruder, B. & Gaussel, N. (2011).** "Risk-Return Analysis of Dynamic Investment Strategies." Lyxor White Paper Series 7. — The continuous-time decomposition of P&L into a convex "option profile" and a running cost. §5.5 derives the discrete analogue.

- **Dao, T.-L., Nguyen, T.-T., Deremble, C., Lempérière, Y., Bouchaud, J.-P. & Potters, M. (2016).** "Tail protection for long investors: Trend convexity at work." *Journal of Investment Strategies.* — Shows that trend P&L equals the difference between long- and short-horizon realised variance. This is the cleanest statement of the central result of this chapter.

4.2 The empirical case

- **Moskowitz, T. J., Ooi, Y. H. & Pedersen, L. H. (2012).** "Time Series Momentum." *Journal of Financial Economics* 104(2), 228–250. — This paper made trend-following academically respectable. It covers 58 futures over 1965–2009 and horizons of one to 12 months, with partial reversal beyond.
- **Hurst, B., Ooi, Y. H. & Pedersen, L. H. (2017).** "A Century of Evidence on Trend-Following Investing." *Journal of Portfolio Management* 44(1), 15–29. — 1880–2016, 67 markets. **[Contested]** AQR sells this strategy, so read it alongside §2.6.
- **Lempérière, Y., Deremble, C., Seager, P., Potters, M. & Bouchaud, J.-P. (2014).** "Two centuries of trend following." *Journal of Investment Strategies.* — This is independent long-history evidence from a different interested party, CFM. Two firms with different books that reach the same conclusion give more evidence than either firm alone.
- **Babu, A., Levine, A., Ooi, Y. H., Pedersen, L. H. & Stamelos, E. (2020).** "Trends Everywhere." *Journal of Investment Management* 18(1), 52–68. — Extends the evidence to less-traded markets and alternative instruments.
- **Hamill, C., Rattray, S. & Van Hemert, O. (2016).** "Trend Following: Equity and Bond Crisis Alpha." Man AHL working paper. — Tests the crisis-alpha claim over a long sample, separated by asset class. **[Contested]** The authors are again an interested party.

4.3 The critiques — read these next to the section above

- **Huang, D., Li, J., Wang, L. & Zhou, G. (2020).** "Time series momentum: Is it there?" *Journal of Financial Economics* 135(3), 774–794. — The drift-confound critique. It determines directly how researchers must specify any new test (§5.3, §10.2).
- **Bhardwaj, G., Gorton, G. B. & Rouwenhorst, K. G. (2014).** "Fooling Some of the People All of the Time: The Inefficient Performance and Persistence of Commodity Trading Advisors." *Review of Financial Studies* 27(11), 3099–3132. — Measures what investors actually received, as opposed to what the strategy produced.
- **Sullivan, R., Timmermann, A. & White, H. (1999).** "Data-Snooping, Technical Trading Rule Performance, and the Bootstrap." *Journal of Finance* 54(5), 1647–1691. — The reality-check methodology, applied to a universe of thousands of technical rules.
- **Park, C.-H. & Irwin, S. H. (2007).** "What Do We Know About the Profitability of Technical Analysis?" *Journal of Economic Surveys* 21(4), 786–826. — The comprehensive survey. It is sober, and useful precisely because its authors have nothing to sell.
- **Kim, A. Y., Tse, Y. & Wald, J. K. (2016).** "Time series momentum and volatility scaling." *Journal of Financial Markets* 30, 103–124. — Argues that much of the reported time-series momentum premium comes from the volatility scaling rather than from the trend signal. The finding is uncomfortable and important (§8.3).

4.4 Implementation and portfolio construction

- **Levine, A. & Pedersen, L. H. (2016).** "Which Trend Is Your Friend?" *Financial Analysts Journal* 72(3), 51–66. — Relates time-series momentum and moving-average rules through their common weighting of past returns.
- **Beekhuizen, P. & Hallerbach, W. G. (2017).** "Uncovering Trend Rules." *Journal of Alternative Investments* 20(2), 28–38. — Derives the return-space weights implied by moving-average rules, and shows that some popular rules have inverted decay or implicit mean reversion. §7.2 builds directly on it.
- **Zakamulin, V. (2017).** *Market Timing with Moving Averages: The Anatomy and Performance of Trading Rules.* Palgrave Macmillan. [paywalled] — A book-length treatment of the same decomposition, plus a large and unusually honest empirical study.
- **Baltas, N. & Kosowski, R. (2013).** "Demystifying Time-Series Momentum Strategies: Volatility Estimators, Trading Rules and Pairwise Correlations." *Journal of Derivatives & Hedge Funds* 19(4), 289–310. — The most directly useful implementation paper in the list. It covers the choice of volatility estimator, turnover reduction, and a correlation-adjusted leverage mechanism.
- **Gârleanu, N. & Pedersen, L. H. (2013).** "Dynamic Trading with Predictable Returns and Transaction Costs." *Journal of Finance* 68(6), 2309–2340. — The closed-form optimal policy with costs. Aim in front of the target, and trade partially toward it. Section 8.7 applies it.
- **Harvey, C. R., Hoyle, E., Korgaonkar, R., Rattray, S., Sargaison, M. & Van Hemert, O. (2018).** "The Impact of Volatility Targeting." *Journal of Portfolio Management* 45(1), 14–33. — What volatility targeting does to the Sharpe ratio, drawdown and skew, by asset class.

- **Carver, R. (2015).** *Systematic Trading.* Harriman House. — The most practically complete public description of a full trend system, from signal scaling through to position rounding. The book takes firm positions and gives specific rules.

4.5 Foundational statistical methods

- **Muth, J. F. (1960).** "Optimal Properties of Exponentially Weighted Forecasts." *Journal of the American Statistical Association* 55(290), 299–306. [paywalled] — The EWMA is the minimum-mean-squared-error forecast for a random walk observed in noise. This explains *why* the exponential kernel is so widely used (§5.8).
- **Lo, A. W. & MacKinlay, A. C. (1988).** "Stock Market Prices Do Not Follow Random Walks: Evidence from a Simple Specification Test." *Review of Financial Studies* 1(1), 41–66. — The variance-ratio test, which §5.4 shows is the same object as a trend rule's expected P&L.
- **Kim, S.-J., Koh, K., Boyd, S. & Gorinevsky, D. (2009).** "ℓ_1 Trend Filtering." *SIAM Review* 51(2), 339–360. — Piecewise-linear trend extraction as a convex program. It formalises the idea that "the trend changed".
- **Kaminski, K. M. & Lo, A. W. (2014).** "When Do Stop-Loss Rules Stop Losses?" *Journal of Financial Markets* 18, 234–254. — Stops help only when returns are positively autocorrelated. Under a random walk, they lower returns. See §11.3.
- **Bailey, D. H. & López de Prado, M. (2014).** "The Deflated Sharpe Ratio." *Journal of Portfolio Management* 40(5), 94–107. — How to discount a Sharpe ratio for the number of trials that produced it.

4.6 If you only read six things

1. **Moskowitz, Ooi & Pedersen (2012)** — the empirical object, stated cleanly.
2. **Huang, Li, Wang & Zhou (2020)** — read immediately after the first, so that you never run an unadjusted test.
3. **Dao et al. (2016)** — what the P&L actually is.
4. **Potters & Bouchaud (2006)** — six pages that permanently correct the intuition about the payoff distribution.
5. **Beekhuizen & Hallerbach (2017)** — which refutes the belief that different moving-average rules are different strategies.
6. **Carver (2015)** — because the papers above leave much work before a working system exists, and this book covers most of that work.

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5. The mathematics of trend-following P&L

This section states exactly what a trend follower earns. It derives four results instead of asserting them:

- §5.2: the expected P&L is a weighted sum of return autocovariances.
- §5.4: for the canonical rule, that sum is *exactly* half the excess of the variance ratio over one. Trend-following and the variance-ratio test are the same statistic.
- §5.5: an exact algebraic identity splits realised P&L into a convex term, a "trend energy" term and a realised-variance cost. It needs no distributional assumptions at all.
- §5.8: the optimal kernel. At realistic signal-to-noise ratios, how long trends last sets the lookback almost entirely, and how strong they are matters little.

5.1 Setup

Fix one market. The trader sets positions from information through t and holds them for one bar, so the P&L over T bars is

$$\Pi_T \;=\; \sum_{t=0}^{T-1} \pi_t \, r_{t+1}, \qquad \pi_t \in \mathcal{F}_t.$$

Two simplifications apply. Both are harmless here, and later sections revisit both. First, π_t is exposure measured in units of capital. Then $\pi_t = 1$ is fully invested, $\pi_t = 2$ is twice levered, and $\pi_t = -1$ is fully short. So $\pi_t r_{t+1}$ is the bar's P&L as a fraction of capital. Costs are absent here and enter in §8.8. Second, assume the signal is a *linear* filter of past returns,

$$s_t \;=\; \sum_{k \geq 0} w_k \, r_{t-k}.$$

§7.2 shows that this covers the moving-average family exactly. The breakout and change-point rules of §6.5 and §6.8 violate it.

5.2 Expected P&L is a weighted sum of autocovariances

Take $\pi_t = s_t$, the linear response with no risk scaling. Assume that returns are covariance-stationary, with mean μ and autocovariances $\gamma_k = \mathrm{Cov}(r_t, r_{t+k})$. Then the expected P&L per bar is

$$\mathbb{E}[\pi_t r_{t+1}] \;=\; \sum_{k\ge 0} w_k\, \mathbb{E}[r_{t-k}\, r_{t+1}] \;=\; \sum_{k\ge 0} w_k(\gamma_{k+1} + \mu^2) \;=\; \underbrace{\sum_{k\ge 0} w_k\, \gamma_{k+1}}_{\text{predictability}} \;+\; \underbrace{\mu^2 \sum_{k\ge 0} w_k}_{\text{drift}}.$$

The second step uses $\mathbb{E}[XY] = \mathrm{Cov}(X,Y) + \mathbb{E}X\,\mathbb{E}Y$, with $\mathbb{E}[r_t] = \mu$ for every t. The lag between r_{t-k} and r_{t+1} is $k+1$, so the index of the autocovariance shifts.

This small identity underlies all of §5. It has three implications.

First, trend-following is long autocovariance. The strategy's edge is a w-weighted inner product with the autocovariance function. If $\gamma_k = 0$ for all $k \ge 1$, as for a martingale difference sequence, the predictability term is exactly zero, whatever the kernel. No filter design extracts edge from an unpredictable series. Any backtest that suggests otherwise is measuring drift, errors in costs, or luck.

Second, the kernel must match the lags where the predictability exists. The inner product $\sum_k w_k \gamma_{k+1}$ is large only when w has mass at the lags where γ does. A slow kernel run on a series whose autocorrelation decays within a day earns nothing. The series is predictable, but the filter weights the wrong lags. A simulation run for this chapter shows this directly. It applied an EWMAC(32, 128) rule, the difference between exponential moving averages of price with spans 32 and 128 (§6.3), to an AR(1) return process with $\phi = 0.03$. The rule earned a Sharpe ratio of 0.00. That kernel's weight is concentrated around lag 30, where $\phi^{30} \approx 2 \times 10^{-46}$. All of the autocovariance lies at lag 1. The kernel's centre of mass must match the decay length of the autocorrelation. This is the one design decision that allows no approximation.

Third, the drift term adds profit that does not come from predictability. A trend kernel has $\sum_k w_k > 0$ by construction. So the drift term $\mu^2 \sum_k w_k$ is *strictly positive* whenever $\mu \neq 0$, whatever the sign of μ. The next subsection covers this.

5.3 The drift confound, stated precisely

Consider the canonical time-series momentum test. It regresses r_{t+1} on $\mathrm{sign}(r_{t-L:t})$, where $r_{t-L:t} = \sum_{i=1}^{L} r_{t-L+i}$ is the cumulative return over the lookback. The estimand is

$$\mathbb{E}\big[r_{t+1} \cdot \mathrm{sign}(r_{t-L:t})\big].$$

Suppose that returns are *completely independent*, with no predictability of any kind, but have mean $\mu > 0$. Write $R = r_{t-L:t}$, which is independent of r_{t+1}. Then

$$\mathbb{E}\big[r_{t+1}\,\mathrm{sign}(R)\big] = \mathbb{E}[r_{t+1}] \cdot \mathbb{E}[\mathrm{sign}(R)] = \mu(2\,\mathbb{P}(R>0) - 1) \;>\; 0$$

because $\mu > 0$ implies $\mathbb{P}(R>0) > 1/2$. **The test statistic is positive under the null.** Take a daily market with $\mu/\sigma = 0.03$ per bar and $L = 252$. Then $\mathbb{P}(R>0) = \Phi(0.03\sqrt{252}) = \Phi(0.476) \approx 0.68$. So the bias is 0.37μ. This is a substantial fraction of the drift itself, and it appears as predictability.

This is the critique of Huang, Li, Wang and Zhou (2020). **[Contested]** Moskowitz, Ooi and Pedersen, and later replications, dispute their conclusion that little time-series momentum survives the correction. The disagreement concerns how to demean: by the full-period in-sample mean, by an expanding window, or cross-sectionally. Nobody disputes the algebra.

What to do about it. Three options exist, in increasing order of conservatism:

1. **Demean each market's returns** with an expanding-window estimate before computing signals and P&L. This is cheap, and it removes most of the bias without look-ahead.
2. **Include the passive long position as a benchmark.** Report trend P&L as an alpha against a constant long position in the same market, at the same average exposure. An alpha of zero means the strategy has rediscovered buy-and-hold.

3. **Test on demeaned bootstrap resamples.** A block bootstrap that preserves the marginal distribution but destroys serial dependence should make a genuine trend effect vanish. If the effect does not vanish, the test statistic measures something other than serial dependence (§10.4).

[Practice] **Recommendation: do all three, and treat any trend result reported without at least the first as uninformative.**

5.4 The exact variance-ratio identity

The next result unifies the practitioner and academic literatures.

The **variance ratio** at horizon q compares the variance of a q-bar return with q times the variance of a one-bar return:

$$\mathrm{VR}(q) \;=\; \frac{\mathrm{Var}\left(\sum_{i=1}^{q} r_i\right)}{q\,\gamma_0} \;=\; 1 + 2\sum_{k=1}^{q-1}\Big(1-\frac{k}{q}\Big)\rho_k.$$

Under a random walk, $\mathrm{VR}(q) = 1$ for all q. Trending series give $\mathrm{VR} > 1$, and mean-reverting series give $\mathrm{VR} < 1$. The variance ratio is the standard non-parametric test for departures from the random walk (Lo and MacKinlay, 1988).

Now take the most-used practitioner rule in existence, **price minus its n-bar moving average**, and compute its expected P&L. Write $\mathrm{MA}_n(p)_t = \frac{1}{n}\sum_{i=0}^{n-1} p_{t-i}$. Then

$$p_t - \mathrm{MA}_n(p)_t \;=\; \frac{1}{n}\sum_{i=0}^{n-1}(p_t - p_{t-i}) \;=\; \frac{1}{n}\sum_{i=0}^{n-1}\sum_{j=0}^{i-1} r_{t-j} \;=\; \sum_{j\ge 0}\frac{(n-1-j)^+}{n}\,r_{t-j},$$

using $(z)^+ = \max(z, 0)$. The last step counts, for each lag j, how many of the n inner sums contain r_{t-j}. So the rule is a linear filter with a **triangular kernel** $w_j = (n-1-j)^+/n$. Substituting into §5.2 with $\mu = 0$ gives

$$\mathbb{E}\Big[\big(p_t - \mathrm{MA}_n(p)_t\big)\,r_{t+1}\Big] = \sum_{j=0}^{n-2}\frac{n-1-j}{n}\,\gamma_{j+1} = \gamma_0\sum_{k=1}^{n-1}\Big(1-\frac{k}{n}\Big)\rho_k = \boxed{\frac{\gamma_0}{2}\Big(\mathrm{VR}(n)-1\Big)}$$

where the last equality is the definition of the variance ratio, rearranged.

This result is exact. The expected per-bar P&L of the price-versus-moving-average rule with window n equals one half of the return variance times the excess of the variance ratio over one at horizon n. A numerical check on simulated AR(1) paths gives an empirical value of 1.630×10^{-5} against a theoretical value of 1.630×10^{-5}.

Three consequences follow:

1. **The trend follower's P&L and the econometrician's random-walk test are the same object.** A firm running a 200-day moving-average rule collects, in expectation, $\frac{1}{2}\sigma^2(\mathrm{VR}(200) - 1)$ per day. To know whether trend-following should work in a market, estimate its variance-ratio profile. No backtest is needed.
2. **The horizon is not a free parameter. It is the horizon of the statistic the strategy is long.** Choosing n means choosing which $\mathrm{VR}(q)$ the strategy is long. VR is typically above one at some horizons and below one at others (§9.2). So the sign of the expected P&L depends on n, independently of the quality of the implementation.
3. **The result explains why trend followers and volatility traders are natural counterparties.** Being long $\mathrm{VR}(n) - 1$ means being long long-horizon variance and short short-horizon variance. Someone selling a spread on the term structure of variance takes the opposite position.

5.5 The exact P&L decomposition: long trend energy, short realised variance

The previous result concerns expectations. This result concerns the realised path. It holds identically for any return sequence whatsoever, with no assumptions about stationarity, distribution or independence.

Let x_t be the EWMA trend estimate, $x_{t+1} = (1-\alpha)x_t + \alpha r_{t+1}$, and take the linear-response position $\pi_t = x_t$. Square the recursion:

$$x_{t+1}^2 = (1-\alpha)^2 x_t^2 + 2\alpha(1-\alpha)\,x_t r_{t+1} + \alpha^2 r_{t+1}^2.$$

Every term except the P&L increment $x_t r_{t+1}$ is a square, so solve for that increment:

$$x_t\, r_{t+1} \;=\; \frac{x_{t+1}^2 \;-\; (1-\alpha)^2 x_t^2 \;-\; \alpha^2 r_{t+1}^2}{2\alpha(1-\alpha)}.$$

Now sum over $t = 0, \ldots, T-1$. The only difficult piece is $\sum_t \left(x_{t+1}^2 - (1-\alpha)^2 x_t^2\right)$, which does not telescope exactly because of the $(1-\alpha)^2$. Split it into a sum that telescopes plus a remainder:

$$x_{t+1}^2 - (1-\alpha)^2 x_t^2 \;=\; \underbrace{\left(x_{t+1}^2 - x_t^2\right)}_{\text{telescopes to } x_T^2 - x_0^2} \;+\; \underbrace{\left(1 - (1-\alpha)^2\right) x_t^2}_{=\,\alpha(2-\alpha)\,x_t^2}.$$

Collecting the three pieces and dividing by $2\alpha(1-\alpha)$ gives the identity

$$\sum_{t=0}^{T-1} x_t\, r_{t+1} \;=\; \underbrace{\frac{x_T^2 - x_0^2}{2\alpha(1-\alpha)}}_{\text{(A) convexity}} \;+\; \underbrace{\frac{2-\alpha}{2(1-\alpha)} \sum_{t=0}^{T-1} x_t^2}_{\text{(B) trend energy}} \;-\; \underbrace{\frac{\alpha}{2(1-\alpha)} \sum_{t=1}^{T} r_t^2}_{\text{(C) realised variance}}$$

A numerical check over 5,000 bars gives $1.1880423066819878 \times 10^{-3}$ for the left side and $1.1880423066819926 \times 10^{-3}$ for the right. The two agree to machine precision, as an algebraic identity must.

Each of the three terms has a distinct interpretation.

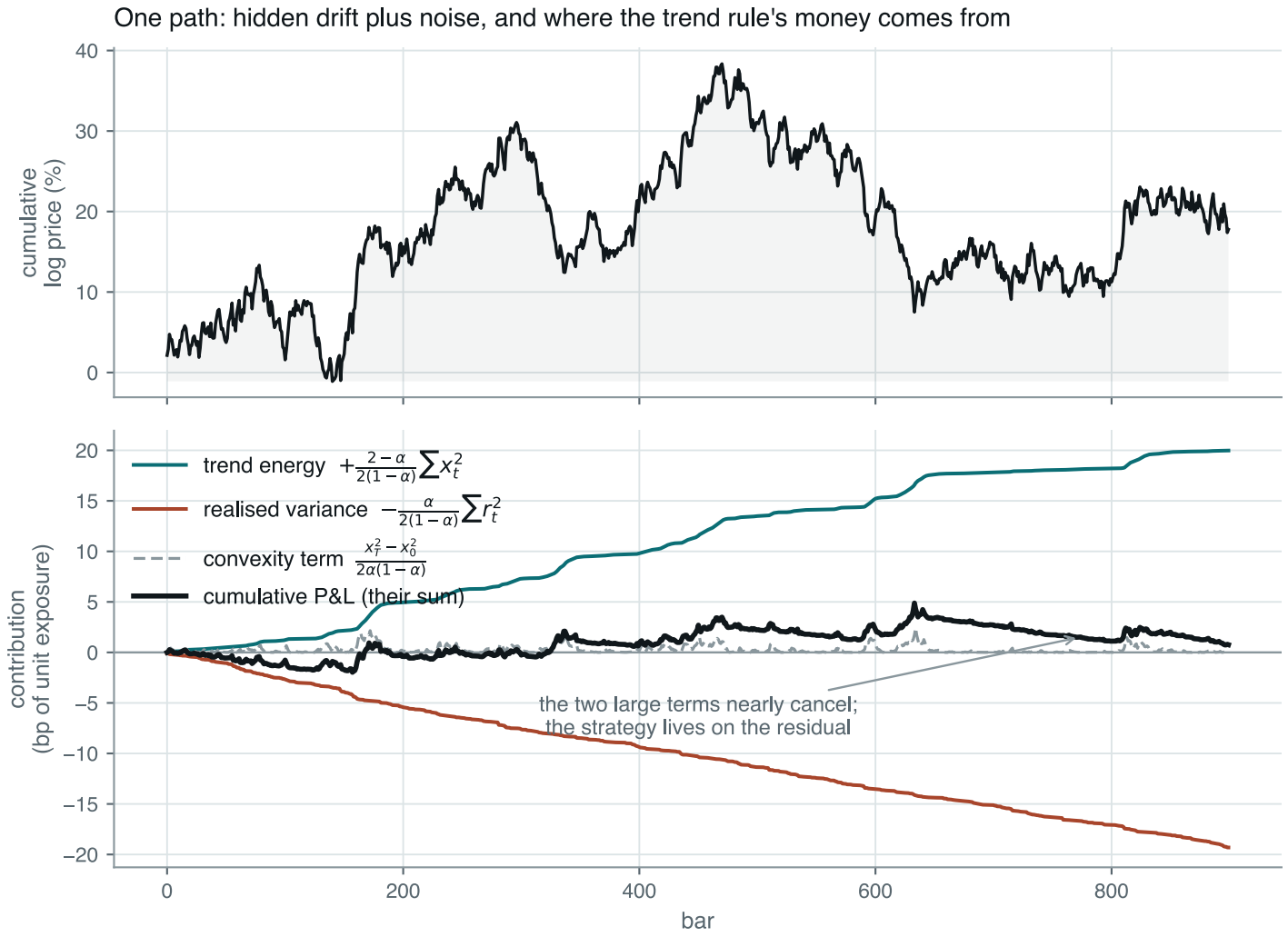
(A) Convexity. This term is the squared trend estimate at the end minus the squared trend estimate at the start. With the usual cold start, $x_0 = 0$, it is non-negative. It is the option payoff. The strategy profits if the trend estimate has moved away from zero, in *either* direction. It does not grow with T, because it is a boundary term. So it contributes nothing to the long-run rate of return. This term pays out exactly in a crisis, when $|x_T|$ is large. This is the algebraic content of "crisis alpha".

(B) Trend energy. This term is the accumulated squared trend estimate. It is always positive, and it grows linearly in T. The strategy collects it while a trend persists. The larger and more sustained the filtered trend, the more this term pays.

(C) Realised variance. This term is the accumulated squared returns, scaled by α, with a minus sign. **This term is the premium.** A faster filter, with larger α, pays more of it. It is the exact, quantified version of "whipsaw".

In summary, **a trend follower is long the energy of the filtered trend and short the realised variance of the underlying.** The speed of the filter fixes the ratio between the two.

This decomposition explains the strategy's behaviour. Under a random walk, the two large terms cancel in expectation. With iid returns of variance σ^2, the steady-state expectation is $\mathbb{E}[x_t^2] = \alpha\sigma^2/(2-\alpha)$. So term (B) contributes $\alpha\sigma^2/(2(1-\alpha))$ per bar, and term (C) subtracts exactly the same amount. The sum is zero, as it must be. Term (A) does not change this. It is $O(1)$ in T, so its contribution *per bar* vanishes. Under a trending process, the drift inflates x_t^2 but not r_t^2, and the residual is the edge. The figure plots the identity on one simulated path.



The two large terms reach roughly $+20$ and -19 basis points of cumulative contribution. The strategy's entire P&L is the residual of about 1 basis point between them. **Trend-following is a small difference between two large numbers. So it is very sensitive to costs, to the volatility estimate and to the speed of the filter**, because all three move one of the two large terms.

Dao and co-authors (2016) reach the equivalent conclusion in continuous time. They state that trend P&L is the difference between long-horizon and short-horizon realised variance. The discrete identity above makes the same statement, with α setting the horizons.

5.6 The option analogy, made precise

Fung and Hsieh (2001) modelled trend followers' returns as a portfolio of **lookback straddles**. These are options that pay the high-minus-low range of the underlying over a period, so the holder receives the best entry and the best exit, chosen after the fact. Fung and Hsieh found that this explained CTA returns far better than linear asset exposures. The analogy is useful. It is exact in some respects and inexact in others. The table compares the two.

Property	Long straddle	Trend rule	Same?
Payoff convex in the underlying's move	Yes	Yes (§1.5 figure)	**Exact**
Premium paid	Once, up front, known	Continuously, as $-\frac{\alpha}{2(1-\alpha)}\sum r_t^2$	Analogous, not identical
Long volatility	Long implied vol	Long *trend* vol, short *realised* vol	**Opposite sign on realised vol**
Payoff depends on path	Only at expiry (vanilla)	Entirely path-dependent	Different

Property	Long straddle	Trend rule	Same?
Maximum loss	Bounded by premium	Unbounded in principle; bounded in practice by sizing	Different
Gap risk	Protected — the option is owned	Exposed — must trade to adjust	**Different, and the most important difference**

The last row matters most. The option protects a straddle holder against a discontinuous move, because its payoff is already contracted. A trend follower's convexity is *synthetic*. The follower creates it by trading, which requires the market to be open and continuous. A gap through the position, such as a limit-locked commodity, a currency de-peg or a weekend, is exactly the state in which the replication fails. **The trend follower is short gap risk and long diffusive convexity.** This differs greatly from a straddle, even though the payoffs on smooth paths match. §11.5 returns to this.

Writers state a second caveat less often. The convexity in the §1.5 figure is convexity **with respect to the underlying's realised move**. It is not convexity with respect to the equity market. Trend-following is long convexity in each of its markets separately. It delivers "crisis alpha" only when a crisis produces a *sustained* directional move that its filters can catch. The 2008 equity decline, which took months, qualified. The February 2018 volatility spike, which took days, did not.

5.7 The trade-level distribution

Combining the convexity of §5.5 with the simulation of §1.5 gives the characteristic shape of trend-following returns. The shape serves as a calibration.

The table comes from a simulation, run for this chapter, of a slow trend rule on a weakly autocorrelated process. The simulation covered 400,000 bars and segmented them into trades at each flip of the position.

Statistic	Value	Interpretation
Hit rate	25%	Three quarters of trades lose
Mean win / mean loss	3.3	Winners are three times the size of losers
Trade P&L skew	$+3.6$	Heavily right-skewed
Contribution of the best 5% of trades	8.2$\times$ the strategy's total net P&L	The rest, in aggregate, lose

The last row matters most. **The best 5% of trades earned more than eight times the strategy's entire net profit, and everything else lost the difference.** This is the design of the strategy, not a pathology. The strategy takes many cheap small losses so that it holds a position during the rare large moves.

Three practical consequences follow:

1. **Analysts cannot evaluate a trend system on a small number of trades.** With this distribution, the sample mean converges very slowly. The effective sample size is closer to the number of *large* moves in the period than to the number of trades.
2. **Any modification that truncates the right tail is likely to eliminate the strategy's edge.** Profit targets are the canonical example. Modifications that truncate the left tail, such as stops, are much less harmful. §11.3 covers that asymmetry.
3. **The reported hit rate is a design choice.** As §1.5 showed, moving from a linear to a sign response raises the fraction of winning *years* from 41% to 50%, with no change in expected return. The same change moves the per-*trade* hit rate in the table above. A manager who advertises a high hit rate is describing their response function.

5.8 What sets the lookback: Muth, Kalman, and the persistence timescale

So far the kernel has been arbitrary. This section finds the right one.

Assume that returns equal a slowly varying hidden drift plus noise. This is the **local level model**. It is the minimal generative model in which trend-following is the correct strategy. The observation is the return, and the hidden state is its drift:

$$r_t = \mu_t + \varepsilon_t, \qquad \mu_t = \phi\,\mu_{t-1} + \eta_t, \qquad \varepsilon_t \sim (0, \sigma_\varepsilon^2),\ \ \eta_t \sim (0, \sigma_\eta^2)$$

Here ϕ is close to 1, so $\tau = 1/(1-\phi)$ is the trend's persistence in bars. Strictly, the local level model is the case $\phi = 1$. In that case the drift is a random walk and never decays. Keeping ϕ just under 1 makes the process stationary and gives

a trend a finite expected lifetime. The analysis needs that finite lifetime. The two parameters that matter are τ and the signal-to-noise ratio $\theta = \mathrm{sd}(\mu_t)/\sigma_\varepsilon$, both measured per bar.

The optimal estimator of μ_t given $\mathcal{F}_t$ is the Kalman filter. For this one-dimensional model, it has a closed form. Its steady-state gain K solves the algebraic Riccati equation. The filter recursion is

$$\hat{\mu}_t \;=\; \phi(1-K)\,\hat{\mu}_{t-1} \;+\; K\,r_t.$$

The recursion is **an EWMA with decay** $\alpha^\star = 1-\phi(1-K)$, up to an overall scale factor. The scale factor arises because the coefficients $\phi(1-K)$ and K sum to 1 only when $\phi = 1$. The scale is irrelevant, because the response function and the risk scaling of §1.3 renormalise the signal anyway. The Kalman filter fixes the *shape* of the kernel. The shape is exponential with decay $\alpha^\star$. When $\phi = 1$, a random-walk trend, the two coefficients do sum to one. The decay then reduces to $\alpha^\star = K$. This recovers the result of Muth (1960). The exponentially weighted moving average is the minimum-mean-squared-error forecast for a random walk observed in noise. This optimality explains why the exponential kernel is so common. The reason is optimality under the simplest model in which trends exist at all, and not convention.

Expanding the product gives the form that matters in practice. The step $1-\phi(1-K) = (1-\phi)+\phi K$ is exact. The only approximation drops the ϕ in front of K. This is legitimate because $\phi \approx 1$:

$$\alpha^\star \;=\; 1-\phi(1-K) \;\approx\; \underbrace{\frac{1}{\tau}}_{\text{trend decay}} \;+\; \underbrace{K}_{\text{signal strength}}, \qquad \text{span} = \frac{2}{\alpha^\star} - 1.$$

The EWMA discounts an old return for exactly two reasons. Either the trend that generated it has decayed ($1/\tau$), or enough fresh evidence has arrived to replace it (K). At realistic parameters, both terms are of order 10^{-3}, so neither is negligible. In the table below, K supplies between a tenth and a half of $\alpha^\star$. The slower the trend, the larger its share.

The range of the two terms, not their size, decides the balance between them. τ varies over orders of magnitude across mechanisms. A trend driven by a policy cycle lasts for quarters. A trend driven by an execution schedule lasts for hours (§2.4). θ, by contrast, is bounded on both sides. If it is too small, no strategy exists. If it is too large, it implies a per-market Sharpe ratio that nobody has achieved. The $\theta = 0.10$ row below implies 1.01 on its own, in one market. So τ decides the lookback, and θ adjusts it.

How long trends last sets the optimal lookback first. How strong they are sets it only second.

The result is useful in practice. [Practice] **Recommendation: do not tune the lookback by maximising backtest P&L.** That objective is noisy and has thousands of effective trials. Instead, estimate the persistence timescale of the process. Use the decay of the autocorrelation, the variance-ratio profile, or the economic mechanism believed to drive the trend. Then set the span to order 2τ. The value 2τ is the weak-signal limit $K \to 0$. The implied spans in the table below range from 0.8τ to 1.7τ. The flatness result that follows shows that this difference costs almost nothing.

The table checks the formula against brute-force optimisation over EWMA spans, on simulated paths of the model above.

τ (bars)	θ	$\alpha^\star$ from Kalman	Implied span	Best span by search	Sharpe at optimum
60	0.05	0.0190	104	101	0.20
120	0.05	0.0105	189	181	0.28
250	0.05	0.0060	333	298	0.36
250	0.10	0.0098	204	214	1.01
500	0.07	0.0048	411	415	0.74

The agreement is close across the range. Compare the third and fourth rows. Holding τ fixed at 250 and *doubling* the signal-to-noise ratio changes the optimal span from 333 to 204. This effect is real, and it shows that the K term matters. But θ cannot vary much in practice. So it adjusts a lookback that τ has already set.

The optimum is flat, and that is the practically important part

Optimality and sensitivity are separate questions. The table sweeps the EWMA span on the process with $\tau = 250$ and $\theta = 0.05$.

Span	5	10	20	40	80	160	320	640
Sharpe	0.085	0.120	0.167	0.221	0.280	0.331	0.352	0.334
% of best	24	34	47	63	79	94	100	95

Being wrong by a factor of two costs about 5% of the Sharpe ratio. Being wrong by a factor of 16 costs half of it. Set the lookback approximately right, and do not optimise it. The data cannot support optimising it (§10.5).

The flat optimum also tests a common practitioner claim. **[Contested]** Practitioners usually justify ensembles across timescales, which average signals over several spans, as adding return. In simulations run for this chapter, ensembles do not add return. An equal-weighted ensemble of six spans from 20 to 640 achieved 93% of the best single span on a stationary process. It achieved 97% on a process whose τ switched among 40, 150 and 600. The ensemble is never better than the best span. Without the ensemble, a wrong choice of span can lose half the Sharpe ratio. With the ensemble, the result is always about 95% of the best. That is insurance against misspecification. Insurance is worth buying, but describe it as insurance, not as alpha.

5.9 Calibration: what edge corresponds to what Sharpe

Putting numbers on the model shows how much predictability a given Sharpe ratio requires. Consider the local level model with $\tau = 250$ days and $\theta = 0.05$, filtered with its optimal EWMA. The table lists the resulting quantities.

Quantity	Value
Lag-1 return autocorrelation ρ_1	≈ 0.0025
VR(60)	≈ 1.14
Per-market annualised Sharpe of the optimal filter	≈ 0.36
t-statistic on ρ_1 from 10 years of daily data	≈ 0.13
Portfolio Sharpe, 50 markets at $\bar{\rho} = 0.15$	≈ 0.88

The first two rows need no simulation. The two parts of $r_t = \mu_t + \varepsilon_t$ are independent, so the lag-k autocorrelation of returns is $\rho_k = \theta^2\phi^k/(1+\theta^2) \approx \theta^2$ for small k. So **the lag-1 autocorrelation equals the squared signal-to-noise ratio**, here $0.05^2 = 0.0025$. Inserting that ρ_k into the variance-ratio formula of §5.4 gives VR$(60) = 1.136$.

The portfolio row uses the standard aggregation identity. Take N equally weighted strategies. Each has the same Sharpe ratio S and the same volatility, and the average pairwise correlation is $\bar{\rho}$. The combined Sharpe ratio is

$$S_{\text{portfolio}} = S\sqrt{\frac{N}{1+(N-1)\bar{\rho}}}.$$

For $N = 50$ and $\bar{\rho} = 0.15$, the multiplier is 2.45.

[Fact] A diversified trend program with a long-run Sharpe ratio near 1.0 corresponds to a daily return autocorrelation of about a quarter of a percent. Ten years of one market's data, $T \approx 2{,}520$ bars, give a standard error of $1/\sqrt{T} \approx 0.02$. So that autocorrelation produces a t-statistic of 0.13. **The effect behind an entire industry is undetectable market by market.** Three conclusions follow:

1. **Market-by-market data cannot validate a trend signal.** A claim that a rule "works on gold" over 10 years reflects noise, whichever sign the result has.
2. **Diversification is the mechanism, not a refinement.** The $\sqrt{N/(1+(N-1)\bar{\rho})}$ multiplier produces the portfolio Sharpe ratio. The per-market signal only has to be marginally positive on average.
3. **The academic difficulty in rejecting the random walk is consistent with the industry's profits.** They are the same fact, observed at different levels of aggregation.

The correlation term matters more than the count. Going from 50 markets to 100 at $\bar{\rho} = 0.15$ improves the multiplier from 2.45 to 2.51, a negligible gain. Reducing $\bar{\rho}$ from 0.15 to 0.05 at $N = 50$ improves it from 2.45 to 3.81, a 55% increase in the Sharpe ratio. **Finding genuinely uncorrelated markets is worth far more than finding more markets.** §8.6 covers this topic in full.

Finally, the calibration implies a required sample size, which governs §10. For iid returns, the standard error of an annualised Sharpe ratio estimated over T years is approximately $\sqrt{(1+S^2/2)/T}$. The $S^2/2$ term is the extra uncertainty from estimating the volatility as well as the mean. The table evaluates this formula.

Sample	$S = 0.4$	$S = 1.0$
5 years	se 0.46 ($t = 0.9$)	se 0.55 ($t = 1.8$)
10 years	se 0.33 ($t = 1.2$)	se 0.39 ($t = 2.6$)
20 years	se 0.23 ($t = 1.7$)	se 0.27 ($t = 3.7$)
50 years	se 0.15 ($t = 2.7$)	se 0.17 ($t = 5.8$)

A 20-year backtest cannot distinguish a Sharpe ratio of 0.4 from zero at conventional significance. This is why the century-long studies of §4.2 exist. It is also why the arguments about them are so hard to settle.

§5 Key takeaways

1. **The expected P&L is $\sum_k w_k \gamma_{k+1} + \mu^2 \sum_k w_k$**: a weighted sum of return autocovariances plus a drift term. Trend-following is long autocovariance and long squared drift.
2. The kernel must put its weight at the lags that carry the autocorrelation. A slow filter applied to fast-decaying autocorrelation earns exactly nothing, however well built.
3. **The drift term biases the standard test positive under the null.** Demean each market before testing, and treat undemeaned trend results as uninformative.
4. **Exact identity:** the price-versus-moving-average rule with window n earns $\frac{1}{2}\gamma_0(\mathrm{VR}(n) - 1)$ per bar. The trend rule and the variance-ratio test are the same statistic, so estimating the VR profile shows whether a backtest is worth running.
5. **Exact identity, with no assumptions:** realised P&L equals a convexity boundary term, plus accumulated trend energy $\sum x_t^2$, minus accumulated realised variance $\sum r_t^2$. The strategy is long trend volatility and short realised volatility.
6. The strategy is a small residual between two large terms. This structure makes it very sensitive to costs and parameters.
7. The straddle analogy is exact for smooth paths and wrong for gaps. The convexity is synthetic, and replicating it requires a tradeable market.
8. **Trend persistence τ, not signal strength, sets the optimal lookback**, with a span of order 2τ. The optimum is also flat. An error by a factor of two costs about 5%.
9. Ensembles across timescales insure against misspecification and do not add return. In simulation, they never beat the best single span.
10. **Calibration to memorise:** a diversified Sharpe ratio near 1.0 corresponds to a daily autocorrelation of about 0.0025, which is undetectable in one market ($t \approx 0.13$ over a decade). Diversification is the mechanism, and lowering $\bar{\rho}$ beats raising N.

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6. Trend estimation and detection

This section catalogues the trend methods. Every method gets the same fields in the same order, so the reader can compare the methods directly. The fields are **intuition, definition, kernel, assumptions, strengths, weaknesses, cost, failure modes, when preferred**. For a linear filter, the section gives the kernel explicitly. §7 shows that the kernel distinguishes the methods.

Six of the ten methods are the same linear filter with different weights, so the catalogue lists fewer than ten different strategies. §5.8 showed that the weights matter far less than the timescale. Read this section for the *failure modes* more than for the definitions. The failure modes differ genuinely, and the definitions mostly do not.

6.1 Lookback return (time-series momentum)

Intuition. Check whether the price is higher than it was L bars ago. If yes, be long.

Definition. $s_t = r_{t-L:t} = p_t - p_{t-L}$, with position $\pi_t = g(s_t)$. The canonical academic form takes $g = \text{sign}$ and $L = 252$.

Kernel. $w_j = \mathbb{1}\{j < L\}$, a rectangle. Every return in the window has equal weight, and every return outside it has zero weight.

Assumptions. The sum of returns over exactly L bars is informative about the next return. Recent returns get no extra weight.

Strengths. It is maximally simple, with no parameters beyond L. Academics state almost all their evidence in this form, so it is the right choice for comparison against the literature.

Weaknesses. The rectangular kernel gives the oldest return in the window exactly the same weight as yesterday's. That is implausible under any model of diffusion. The kernel also gives the return just before the window zero weight. This discontinuity has no economic meaning. So the signal jumps when a large old return leaves the window. The jump is an artefact and carries no information.

Cost. $O(1)$ per bar, from the running sum.

Failure modes.

- (i) The drop-out artefact above generates trades on no news.
- (ii) It is maximally exposed to the drift confound of §5.3, because sign$(p_t - p_{t-L})$ is positive most of the time in a drifting market.
- (iii) With $g =$ sign, it discards the magnitude of the signal, and with it the convexity of §1.5.

When preferred. For benchmarking and communication. [Practice] **Recommendation: do not run it in production.**

6.2 Price versus moving average

Intuition. Check whether the price is above its own recent average.

Definition. $s_t = p_t - \text{MA}_n(p)_t$, where MA is a simple or exponential average over n bars.

Kernel. For the simple average, $w_j = (n-1-j)^+/n$, a **descending ramp** from $(n-1)/n$ at lag 0 down to zero at lag $n-1$. For the exponential average, $w_j = (1-\alpha)^{j+1}$ with $\alpha = 2/(n+1)$, so its span is also n. This kernel is a clean exponential decay. §5.4 and §7.2 derive both kernels.

Assumptions. Recent returns matter more than old ones, with weights that decay linearly (SMA) or geometrically (EWMA).

Strengths. The decay is economically sensible, and the EWMA version has no drop-out artefact. By §5.4, on demeaned returns, its expected P&L is exactly $\frac{1}{2}\gamma_0(\text{VR}(n) - 1)$. That makes it the most analytically tractable rule in the catalogue.

Weaknesses. The rule puts its maximum weight on the most recent return. That return is the noisiest one. Microstructure effects such as bid-ask bounce and closing auctions contaminate it most. So the raw signal is erratic, and turnover is high.

Cost. $O(1)$ per bar for the EWMA, and $O(1)$ with a running sum for the SMA.

Failure modes.

- (i) In a market that mean-reverts quickly, the heavy weight on lag 0 makes the rule a *reversal* signal with the wrong sign.
- (ii) It carries no information about scale. So the raw s_t is not comparable across markets or across volatility regimes, and it must be normalised (§8.3). Forgetting to normalise is the most common implementation bug in this whole catalogue.

When preferred. As the default. [Practice] **Recommendation: if only one trend rule is used, use this one, in its EWMA form, normalised by volatility.**

6.3 Moving-average crossover

Intuition. Check whether the short-term average is above the long-term average. Equivalently, check whether the recent past is better than the more distant past.

Definition. $s_t = \text{MA}_m(p)_t - \text{MA}_n(p)_t$, with $m < n$. The EWMA version ("EWMAC") with spans m and n is the industry standard.

Kernel. $w_j = \dfrac{(n-1-j)^+}{n} - \dfrac{(m-1-j)^+}{m}$. As §7.2 shows, it rises linearly to a peak at lag $m-1$, then decays linearly to zero at lag $n-1$. This kernel is a **band-pass filter** that deliberately *down-weights the most recent returns*.

Assumptions. The informative frequency band is bounded away from both zero and the Nyquist frequency. The very recent past is noise, and the very distant past is stale, so the rule uses neither.

Strengths. Down-weighting recent returns removes exactly the microstructure noise that hurts the rule of §6.2. The result is materially lower turnover for a similar gross edge. That is why most production systems run this rule rather than the simpler one.

Weaknesses. It has two parameters instead of one, and their ratio interacts with the timescale. This interaction is easy to overfit. Suppressing recent returns introduces lag, which is costly at turning points.

Cost. $O(1)$ per bar.

Failure modes.

- (i) Certain (m, n) combinations produce kernels with a *negative* lobe. The lobe is an implicit mean-reversion component that the designer did not intend. Beekhuizen and Hallerbach (2017) document this for common multi-average rules. The lobe is invisible in price space and obvious in return space.
- (ii) In choppy markets, the signal crosses zero frequently. With a sign response, that generates a burst of loss-making trades, the classic whipsaw.

When preferred. In production, when turnover matters. [Practice] **Recommendation: use spans in a ratio of roughly 1:4, and set the slow span from τ as in §5.8.**

6.4 Regression slope

Intuition. Fit a straight line to the recent log price, and use its slope.

Definition. $(\hat{a}_t, \hat{\beta}_t) = \arg\min_{a,\beta} \sum_{i=0}^{L-1} \left(p_{t-i} - a - \beta\,(t-i)\right)^2$, and the signal is the fitted slope, $s_t = \hat{\beta}_t$. The intercept is a nuisance parameter, fitted and discarded. A common variant divides by the standard error of the slope. That gives a t-statistic, which normalises for volatility automatically.

Kernel. $w_j = \dfrac{6(j+1)(L-1-j)}{L(L^2-1)}$, a **centred inverted parabola**. It is symmetric, peaks in the middle of the window and vanishes at both ends. A check against direct computation confirms this closed form. The weights sum to one, so a path that is a pure straight line returns its own slope.

Assumptions. The log price is locally linear plus independent noise, and the noise is homoskedastic within the window.

Strengths. The t-statistic variant normalises itself. The symmetric kernel is maximally smooth, which gives a very stable signal.

Weaknesses. The symmetry also causes the main weakness. The kernel weights the middle of the window most, so its centre of mass is at $(L-2)/2$ and the estimate lags by $\approx L/2$ bars. That lag is half again as long as the lag of price-versus-SMA(L), whose centre of mass is at $(L-2)/3$. The L-bar lookback return of §6.1 has the same lag of $\approx L/2$. Even so, the parabola reacts more slowly to a fresh move than the rectangle does. The rectangle gives yesterday's return a $1/L$ share of the total weight. The parabola gives it only $6/(L(L+1))$. No other method in the catalogue reacts more slowly. The rule rejects noise entirely at the cost of timeliness.

Cost. $O(1)$ per bar, with running sums of p, tp, t and t^2.

Failure modes.

- (i) The lag makes it late to reversals, so it gives back more at turning points than the exponential rules.
- (ii) The t-statistic version divides by a residual standard error estimated inside the window. For short L, that estimate is *very* noisy. It produces enormous spikes in the signal when a window happens to be quiet. Use a separate, longer volatility estimate instead.

When preferred. When the stability of the signal matters more than responsiveness, and in research settings where the interpretability of the t-statistic helps.

6.5 Breakout and channel rules

Intuition. Check whether the price has exceeded its highest level of the last L bars.

Definition. The Donchian rule goes long when $P_t > \max_{1\le i\le L} P_{t-i}$, goes short when $P_t < \min_{1\le i\le L} P_{t-i}$, and holds otherwise. Variants add a shorter exit channel, or a band scaled by volatility around a moving average (Keltner, Bollinger).

Kernel. None. This is not a linear filter. The signal is a function of order statistics of the price path. So the analysis of §5.2 does not apply, and no weighted sum of autocovariances gives the expected P&L.

Assumptions. Extremes carry information beyond what the mean of the window conveys. The *maximum* carries information that the *average* does not.

Strengths. The rule is naturally discrete. It produces few, well-separated trades, and therefore low turnover. It is insensitive to the shape of the return distribution within the channel. Its economic rationale is also interpretable: resting stop orders cluster above the range.

Weaknesses. It discards almost all the data, because only the extremes matter. It is also discontinuous, so a one-tick difference flips the position.

Cost. $O(1)$ amortised per bar with a monotonic deque, $O(\log L)$ with a heap or an ordered multiset, and $O(L)$ naively.

Failure modes.

- (i) The extreme of a window is a maximally noisy statistic. A single bad print or a stale quote creates a false breakout, so data cleaning matters more for this rule than for any other.
- (ii) In markets with limit moves or frequent gaps, the price often jumps past the breakout level instead of touching it. So the realised entry differs systematically from the backtested one. The backtests of this rule are optimistic in exactly the markets where trend-following earns its return.
- (iii) Being flat inside the channel makes the position a step function of the price. This makes risk at the portfolio level uneven.

When preferred. When costs dominate and a genuinely low-turnover system is needed. It also serves as a diversifier alongside a linear filter, since its errors are uncorrelated with the filter's.

6.6 State-space models and the Kalman filter

Intuition. Posit that an unobserved trend *does* exist, and estimate it optimally.

Definition. Use the local level model of §5.8, or its local *linear* trend extension, in which both a level and a slope evolve:

$$\begin{aligned} p_t &= \ell_t + \varepsilon_t \\ \ell_t &= \ell_{t-1} + b_{t-1} + \xi_t \\ b_t &= \phi\, b_{t-1} + \eta_t \end{aligned}$$

Here ℓ_t is the unobserved level, b_t the trend, and ε_t, ξ_t and η_t are mutually independent zero-mean innovations. This model is written in *price* space. So ε_t here is observation noise on the log price. It differs from the return noise that carries the same name in §5.8. The Kalman filter delivers both $\mathbb{E}[b_t \mid \mathcal{F}_t]$ and its variance.

Kernel. For the one-state model of §5.8, the steady-state filter is exactly an EWMA with $\alpha^\star = 1 - \phi(1 - K)$. So it is the exponential rule with a parameter derived from a model, and it is not a new estimator. For the two-state model above, the filter is a fixed linear combination of two exponential smoothers, one for the level and one for the slope. This combination is Holt's method with damping. It adds one parameter to the one-state case but is still built from exponential smoothers. This equivalence is the most useful point in this subsection.

Assumptions. Linear Gaussian dynamics with known variances. Financial returns badly violate the Gaussian part. The linearity is acceptable.

Strengths. The filter gives the *uncertainty* of the trend estimate as well as its value. No other method here does this. The uncertainty enables position sizing scaled by $\widehat{b}_t / \operatorname{sd}(\widehat{b}_t)$ and valid confidence intervals. The filter also handles missing data and irregular sampling directly. This matters for a global futures panel with holidays that do not coincide.

Weaknesses. Estimating the innovation variances from data is a badly conditioned problem at realistic signal-to-noise ratios. The quantity to estimate is the ratio of state variance to observation variance. This ratio is the θ^2 of §5.8, of order 10^{-3} at the calibration of §5.9. At that value the likelihood is nearly flat, so maximum-likelihood estimates are unstable. In practice, users set the parameters by judgement. This forfeits most of the method's claimed advantage.

Cost. $O(d^3)$ per bar for a d-dimensional state, from propagating the covariance. With $d = 2$ or 3, the cost is negligible.

Failure modes.

- (i) Fitting the variances by maximum likelihood on a short sample typically ends at a corner solution. Either $\widehat{\sigma}_\eta \approx 0$, which means no trend ever, or the filter is very fast.

- (ii) Under the Gaussian assumption, the filter treats a single large return as strong evidence of a change in trend, rather than as a draw from a fat tail. So it over-reacts to jumps. A Student-t observation model, or simple winsorisation, fixes this and is worth using.

When preferred. When uncertainty estimates are needed, when the data are irregular, or when a lookback should be justified from a model rather than a backtest.

6.7 Smoothers: Hodrick–Prescott, ℓ_1, and wavelets

Intuition. Separate the price into a smooth component and a rough one, and trade the direction of the smooth one.

Definition. Each of these methods solves a penalised least-squares problem over the whole path. Hodrick–Prescott picks the smooth component $\{\ell_t\}_{t=1}^{T}$ that minimises

$$\sum_{t=1}^{T}(p_t-\ell_t)^2 \;+\; \lambda_{\mathsf{HP}}\sum_{t=2}^{T-1}\left((\ell_{t+1}-\ell_t)-(\ell_t-\ell_{t-1})\right)^2.$$

The first sum measures fit to the observed path, and the second penalises curvature. So λ_{HP} sets the weight on smoothness relative to fit. ℓ_t is the same kind of object as the level in §6.6. ℓ_1 trend filtering (Kim, Koh, Boyd and Gorinevsky, 2009) replaces the squared second differences with absolute ones. This gives a *piecewise linear* trend with a small number of kinks. The kinks formalise the statement "the trend changed here". Wavelet methods decompose the path into scales and reconstruct it from the coarse scales.

Kernel. For HP, away from the ends of the sample, the kernel is symmetric and two-sided. The estimate at t weights observations *after* t exactly as heavily as observations before it. This use of future data is the central issue.

Assumptions. A smoothness penalty reflects a true property of the process that generates prices.

Strengths. ℓ_1 trend filtering in particular produces straight-line trend segments joined at breakpoints. Practitioners describe trends informally in exactly this way. The formulation is convex, so a solver can find the global solution, and it has one interpretable parameter.

Weaknesses. These are smoothers, not filters. They use future data to estimate the trend at time t. Their excellent in-sample fit is therefore unavailable in real time.

Cost. HP costs $O(T)$ for the whole path, via a banded solve. ℓ_1 is a convex program, at $O(T)$ per interior-point iteration.

Failure modes.

- (i) **The look-ahead error.** Applying HP or ℓ_1 to the full sample and then backtesting on the extracted trend produces spectacular, entirely fictitious results. [Practice] This is the most common serious error in trend research. The only correct use re-solves the problem on the expanding window $\{p_1,\dots,p_t\}$ at every t. This is expensive. It also degrades the estimate at the right-hand edge, which is where the estimate is needed.
- (ii) Even when used correctly, the endpoint estimate of a two-sided smoother is far noisier than the interior one, so the apparent smoothness of the fitted path is misleading.

When preferred. For *analysis*, such as labelling historical regimes, defining trend episodes for study, and generating targets for supervised learning. They are rarely right for live signals. A one-sided filter is the appropriate choice there.

6.8 Change-point detection

Intuition. Instead of estimating a trend continuously, detect the moments when the regime changes.

Definition. The classical CUSUM statistic accumulates deviations from a reference. It signals when the accumulation exceeds a threshold. The statistic is $S_t = \max(0,\; S_{t-1}+r_t-\delta)$, with an alarm when $S_t > h$. Here δ is a slack term, set to about half the drift that the test should detect. The slack keeps a series without drift near zero. h is the alarm threshold. This one-sided form detects upward shifts. A mirrored copy run on $-r_t$ detects downward shifts. Bayesian online change-point detection instead maintains a posterior over the *run length*, the time elapsed since the last change.

Kernel. Nonlinear and adaptive. The effective lookback resets at each detection, which is exactly the property that a filter with a fixed kernel lacks.

Assumptions. Regimes are genuinely discrete, and the data allow estimating the distributions before and after a change.

Strengths. CUSUM has a classical guarantee of optimal detection delay for a given rate of false alarms. It also provides an adaptive lookback at no extra cost. Combining change-point detection with a trend filter is one of the more promising current research directions. The change point shows when the filter should discard older data.

Weaknesses. The distributions before and after a change are not known, and the analyst must estimate them from very few observations near the change. The assumption that regimes are discrete is also a modelling convenience. It is not a fact about markets.

Cost. $O(1)$ per bar for CUSUM. $O(T)$ per bar naively for Bayesian online detection, and $O(1)$ amortised with pruning.

Failure modes.

- (i) The threshold h sets the trade-off between false alarms and detection delay. At trend-following's signal-to-noise ratios (§5.9), the operating point is poor. A tolerable rate of false alarms comes with a delay comparable to the trend's own duration.
- (ii) Detections cluster during volatile periods. That produces the worst behaviour. The signal changes most when markets are most expensive to trade.

When preferred. As a *modifier* on a filter, shortening the effective lookback after a detection, rather than as a standalone signal.

6.9 Persistence statistics

Intuition. Instead of estimating the direction of a trend, estimate whether this market is the kind of market that trends at all.

Definition. Three statistics are in common use:

- **The variance ratio** $\mathrm{VR}(q)$, as defined in §5.4. A value above one means trending. By §5.4 it is also more than a diagnostic. The quantity $\frac{1}{2}\gamma_0(\mathrm{VR}(q) - 1)$ equals the expected per-bar P&L of the price-versus-SMA(q) rule. So estimating the statistic and backtesting that rule are the same calculation.
- **The Hurst exponent** H, from the scaling $\mathrm{Var}(p_{t+q} - p_t) \propto q^{2H}$. $H = 1/2$ is a random walk, and $H > 1/2$ is persistence. Since $\mathrm{VR}(q) \propto q^{2H-1}$, H and the slope of the VR profile carry the same information.
- **The efficiency ratio** (Kaufman): $|p_t - p_{t-L}| \big/ \sum_{i=0}^{L-1} |r_{t-i}|$. It is net progress divided by gross path length, and the triangle inequality puts it in $[0, 1]$. A monotone path gives 1. A random walk gives about $1/\sqrt{L}$. The numerator grows like $\sqrt{L}$, the denominator grows like L, and the factor $\mathbb{E}|r|$ common to both cancels. In simulation, the mean ratio at $L = 100$ is 0.100.

Assumptions. Stationarity over the estimation window. That is a strong assumption for a statistic whose whole purpose is to detect differences between regimes.

Strengths. These are *conditioning* variables, not signals. They indicate how far to trust a trend signal, which markets to include, or when to reduce size. Used that way, they are among the most defensible tools here.

Weaknesses. All three are noisy. H in particular requires far more data than practitioners typically use. The standard estimators, rescaled range in particular, have severe small-sample bias. A random walk of a few hundred points routinely yields $\hat{H} \approx 0.6$.

Cost. $O(L)$ to $O(L \log L)$ per update.

Failure modes.

- (i) Users estimate H or VR on short windows and treat the result as a regime classification. The noise of the estimator dominates the result.
- (ii) Circularity. Selecting markets by their historical variance ratio, and then reporting the trend performance of the selected set, is plain selection bias, and it is common.

When preferred. For selecting markets at long horizons, and for research diagnostics. Not as a real-time regime switch.

6.10 Machine-learned signals

Intuition. Learn the map from past returns to position directly, instead of specifying it.

Definition. Lim, Zohren and Roberts (2019) train an LSTM to output a position directly, by maximising a Sharpe-ratio objective. Its inputs are volatility-normalised returns and a set of standard trend signals. Later work replaces the recurrent architecture with attention and adds change-point features.

Kernel. Learned, nonlinear and time-varying. This is the method's main advantage and its main risk.

Assumptions. The data contain enough signal, relative to the number of effective parameters, to identify a richer function than a linear filter.

Strengths. The model can learn the response function g and the kernel jointly. It can condition on state, such as volatility, correlation and positioning. The papers report real improvements over a classical rule.

Weaknesses. The signal-to-noise calibration of §5.9 limits the method. With a per-market daily information coefficient of order 0.01, and a few thousand effectively independent observations, the data support estimating a handful of parameters and not thousands. **[Contested]** The published improvements are genuine on their samples. Whether they survive as out-of-sample premia is not established.

Cost. The cost of training is irrelevant. The relevant cost is the number of research trials, which §10.5 penalises.

Failure modes.

- (i) The model may learn the sample's specific volatility regime instead of a trend relationship. Researchers typically train on volatility-scaled returns to prevent this. Verify that the prevention worked.
- (ii) Silent look-ahead through the normalisation. Computing the volatility scaling or the feature standardisation on the full sample leaks the future, and in a low-signal setting the leak is bigger than the signal.

When preferred. With a genuinely large cross-section, a disciplined out-of-sample protocol, and a specific hypothesis about the nonlinearity or conditioning expected. Not as a first system.

6.11 Comparison

The table compares the ten methods on the attributes that distinguish them. The ratings are this chapter's assessment. They rest on the calibration of §5.9 and not on any single dataset.

Method	Linear?	Params	Turnover	Lag	Robustness	Main failure mode
Lookback return	Yes	1	Medium	Medium	Medium	Drift confound; window drop-out
Price vs MA	Yes	1	High	Low	Medium	Overweights noisiest lag
MA crossover	Yes	2	Medium	Medium	Good	Unintended negative kernel lobe
Regression slope	Yes	1	Low	**High**	Good	Late at turning points
Breakout	No	1–2	**Low**	Medium	Medium	Gap-through; optimistic backtests
Kalman / state space	Yes†	2–3	Medium	Low	Medium	Unstable variance estimation
HP / ℓ_1 smoothers	Yes‡	1	Low	—	Poor§	Look-ahead if misused
Change-point	No	2	Bursty	**High**¶	Poor	Bad delay/false-alarm trade-off
Persistence statistics	No	1–2	—	—	Poor	Estimator noise; selection bias
Machine-learned	No	Many	Variable	Variable	Unknown	Over-parameterisation; leakage

† Linear in steady state, because it is an EWMA (§5.8). ‡ Linear but two-sided, hence not a causal filter. § Robustness refers to real-time use. As offline analysis tools these are fine. ¶ Detection delay, comparable to the trend's own duration at these signal-to-noise ratios (§6.8). The lag *after* an alarm is short, since the effective lookback resets.

§6 Key takeaways

1. Six of these ten methods are the same linear filter with different weights. Read the catalogue for failure modes, not for definitions.
2. **Price versus an exponential moving average, normalised by volatility, is the right default.** Every other choice needs a reason.
3. The crossover's real advantage is that it *down-weights the most recent returns*, which cuts microstructure noise and turnover. Its advantage does not come from comparing a faster average with a slower one.
4. The regression slope's symmetric kernel gives smoothness at the cost of a lag of about half the window. That trade-off is sometimes right, and it is always explicit.
5. Breakouts are not linear filters, and the analysis of §5 does not cover them. Their backtests are systematically optimistic in gapping markets.
6. The Kalman filter is an EWMA with a decay derived from a model. Its main contribution is the *uncertainty estimate*, which enables sizing that reflects uncertainty.
7. **HP and ℓ_1 smoothers are two-sided.** Using them without re-solving on an expanding window is the most common serious look-ahead error in trend research.
8. Persistence statistics belong in market selection and diagnostics, not in real-time switching. At any window short enough to be responsive, the noise of the estimator exceeds the effect.

7. Taxonomy and equivalences

7.1 The master form, with slots

Write the pipeline of §1.3 as a design space instead of a single formula:

$$\pi_{i,t} \;=\; \underbrace{c_t}_{\text{portfolio}} \cdot \underbrace{\frac{\sigma^\star}{\widehat{\sigma}_{i,t}}}_{\text{risk}} \cdot \underbrace{g}_{\text{response}} \Big(\underbrace{\textstyle\sum_k w_k \, \tilde{r}_{i,t-k}}_{\text{kernel}} \Big)$$

As in §1.3, $\tilde{r}_{i,t}$ is market i's return after whatever normalisation the signal stage applies, in practice $r_{i,t}/\widehat{\sigma}_{i,t}$. c_t is one multiplier at the portfolio level, common to every market. The volatility estimate appears both in the kernel's input and in the risk slot. This repetition is deliberate and is not double-counting. §8.9 explains why. The table describes the four slots.

Slot	What it controls	Choices
$\{w_k\}$	**Which horizon** is bet on, and how the past is weighted	box, ramp, exponential, hump, parabola, learned
$g(\cdot)$	The **shape of the payoff** and the turnover	linear, capped-linear, sign, bump-shaped, thresholded
$\widehat{\sigma}_{i,t}$	Risk comparability across markets and time	EWMA of squared returns, range estimators, blends of horizons
c_t	Total portfolio risk and its dynamics	fixed target, correlation-adjusted, drawdown-modulated

The slots are close to orthogonal, which is the practical value of this form. The kernel can change without changing the response, and the volatility estimator can change without changing either. So the slots can be ranked by the effort they deserve, as §7.3 does.

7.2 The kernel table: exact equivalences

Every linear rule in §6 is a choice of $\{w_k\}$, and the weights have closed forms. The table below is the most useful half-page in the chapter. It reduces many named "indicators" to one object, the kernel.

Rule	Weight w_j on r_{t-j}	Shape	Peak at lag	Centre of mass
L-bar lookback return $p_t - p_{t-L}$	$\mathbb{1}\{j < L\}$	box	flat	$(L-1)/2$
Price $-$ SMA(n)	$\dfrac{(n-1-j)^+}{n}$	descending ramp	0	$(n-2)/3$
Price $-$ EWMA(α)	$(1-\alpha)^{j+1}$	exponential	0	$(1-\alpha)/\alpha$
SMA(m) $-$ SMA(n), $m < n$	$\dfrac{(n-1-j)^+}{n} - \dfrac{(m-1-j)^+}{m}$	tent / band-pass	$m-1$	$\dfrac{m+n-3}{3}$
EWMAC(α_f, α_s)	$(1-\alpha_s)^{j+1} - (1-\alpha_f)^{j+1}$	hump / band-pass	$\approx \dfrac{\ln(\alpha_f/\alpha_s)}{\alpha_f - \alpha_s} - 1$	$\dfrac{1-\alpha_s}{\alpha_s} + \dfrac{1-\alpha_f}{\alpha_f}$
OLS slope over L bars	$\dfrac{6(j+1)(L-1-j)}{L(L^2-1)}$	centred parabola	$(L-2)/2$	$(L-2)/2$
Kalman, local level, steady state	$\propto (1-\alpha^\star)^j$, $\alpha^\star = 1-\phi(1-K)$	exponential	0	$(1-\alpha^\star)/\alpha^\star$

Every weight and every centre of mass in the table is an algebraic identity, not an approximation. Each was checked numerically against direct computation on simulated paths. The single approximate entry is the peak of the EWMAC, which treats the lag as continuous. It is accurate to about a bar. §5.4 and §6.4 derive the ramp and the parabola. The same telescoping argument gives the exponential row and the two band-pass rows. The last column shows a notable property of the two band-pass rules. Each one's centre of mass is the *sum* of the centres of mass of the two price-versus-average rules whose difference it is (for the SMA pair, up to a $1/3$). A crossover therefore reaches further back than either of its own components. The figure draws five of the kernels on a common lag axis.

Every measure is a weighted sum of past returns. Only the kernel differs.

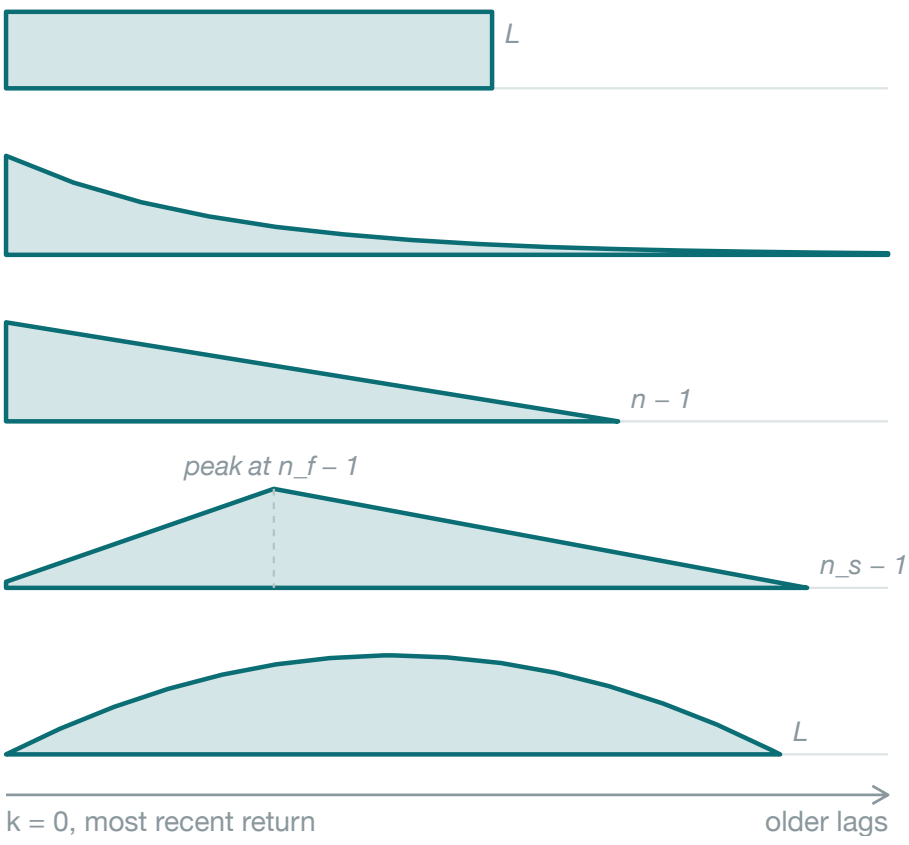
Lookback return
rectangular
`w_k = 1/L, k < L`

EWMA
exponential
`w_k ∝ λ^k`

MA displacement
descending ramp
`w_k ∝ n−1−k`

MA crossover
hump / band-pass
`fast − slow`

Regression slope
centred parabola
`w_k ∝ (k+1)(L−1−k)`



Four readings of the table change how a system should be designed.

A moving-average crossover down-weights recent returns. The hump peaks at an interior lag. So yesterday's return, the noisiest observation available, gets less weight than the return at the hump's peak, $m-1$ bars back. That is a feature. It is the actual reason crossovers beat the simpler rule net of costs. Most descriptions of the crossover, as "fast average versus slow average", omit this property completely.

Some popular rules contain hidden mean reversion. A signal built from several moving averages with badly chosen windows can have a kernel that goes negative at some lags. The rule then takes a position *against* returns from that part of the past. Beekhuizen and Hallerbach (2017) document this for common multi-average constructions. It is invisible in price space. The remedy takes one line of code. Expand the rule into return weights and plot them. If any $w_j < 0$ unintentionally, fix the rule.

The lookback of a rule has a precise meaning. Two rules with different names and similar centres of mass are nearly identical strategies. Two rules with the same name and different windows are not. Compare rules by the centres of mass of their kernels, not by their parameter labels. A price-versus-SMA(200) rule has its centre of mass at $(200-2)/3 \approx 66$ bars. A price-versus-EWMA rule of span 200 has it at $(1-\alpha)/\alpha \approx 100$ bars. Both are nominally a "200-day rule", yet their effective horizons differ by 50%.

The regression slope is the slowest common rule to react. Its centre of mass is at $(L-2)/2$, against $(n-2)/3$ for the moving-average rule. For the same nominal window, it looks 50% further into the past. The L-bar lookback return sits at the same $\approx L/2$, but it gives yesterday's return full weight, where the parabola gives it almost none. Smoothness costs timeliness, and the centre-of-mass column measures that cost.

7.3 Which slot matters

The taxonomy yields a ranking of the slots. The table ranks them by the size of the effect in simulation and by what the literature supports.

Rank	Slot	Effect size	Evidence
1	**Risk scaling** $\widehat{\sigma}_{i,t}$	Sharpe 0.04 $\rightarrow$ 0.36 in this chapter's simulation with volatility clustering	Kim, Tse & Wald (2016) attribute much of the reported premium to it; Harvey et al. (2018) document its effect on drawdowns
2	**Portfolio aggregation** c_t	Sharpe $\times 2.45$ for 50 markets at $\bar{\rho} = 0.15$ (§5.9)	Arithmetic; Baltas & Kosowski (2013) on correlation adjustment
3	**Kernel timescale**	\$±\$5% for a factor-of-two error, \$-\$50% for a factor of sixteen (§5.8)	This chapter's simulation; consistent with the flatness reported across the literature
4	**Response function** g	Changes skew and hit rate substantially; changes Sharpe modestly	§1.5; Lempérière et al. (2014) find saturation for large signals
5	**Kernel shape** at fixed timescale	Small	Levine & Pedersen (2016); Beekhuizen & Hallerbach (2017)

The ordering is close to the inverse of the attention these slots get in the practitioner literature. That literature is dominated by discussion of indicators (rank 5) and lookbacks (rank 3). [Practice] **Recommendation: given one week to improve a trend system, spend it on the volatility estimator and the correlation structure.**

Rank 1 comes with a caveat. Volatility scaling does not improve the *signal*. It helps for two reasons. (i) It makes risk comparable across markets, so the instrument that is currently most volatile does not dominate the portfolio. (ii) Volatility is far more predictable than returns, so dividing by a forecastable quantity improves the resulting ratio. **[Contested]** Kim, Tse and Wald (2016) go further. They argue that the volatility scaling, and not the trend signal, produces much of the reported time-series momentum premium. If they are right, a large part of what the industry calls trend-following is really a volatility-managed long position. That would be a different product, with a different fee.

7.4 Same thing, different names; same name, different things

The table records which common identifications are exact and which are false.

Claim	Status
Price-vs-MA(n) rule $\equiv \frac{1}{n}\sum_{L=1}^{n-1}$ of the L-bar momenta	**Exact** (§5.4)
Its expected P&L $\equiv \frac{1}{2}\gamma_0(\text{VR}(n)-1)$	**Exact** on demeaned returns; a drifting market adds $\mu^2\frac{n-1}{2}$ (§5.2)
Kalman filter on the local level model $\equiv$ EWMA with $\alpha^\star = 1-\phi(1-K)$	**Exact** in steady state (§5.8)
Hurst exponent $H \equiv \frac{1}{2}(1+$ slope of $\log \text{VR}(q)$ against $\log q)$	**Exact** given the scaling assumption ($\text{VR}(q) \propto q^{2H-1}$)
MACD $\equiv$ EWMAC	**Exact** — MACD *is* an EWMA crossover, usually spans 12 and 26
"Trend-following" $\equiv$ "time-series momentum"	**Approximate.** TSMOM is the sign-response, box-kernel special case
"Trend-following" $\equiv$ "managed futures"	**False.** One is a strategy, the other an industry with fees
"Momentum" (cross-sectional) $\equiv$ "momentum" (time-series)	**False and consequential.** Different signals, different portfolios, different risks — see Goyal and Jegadeesh's decomposition
Bollinger-band breakout $\equiv$ Donchian breakout	**False.** One is a volatility-scaled band around a mean, the other an order statistic
Volatility targeting $\equiv$ risk parity	**False.** The first sets total risk over time, the second allocates risk across assets

The cross-sectional versus time-series momentum row needs attention, because the two strategies share a word and are routinely conflated. A cross-sectional momentum portfolio is dollar-neutral by construction. A time-series momentum portfolio has a net directional position, which varies with how many markets are trending. That net position causes most of trend-following's behaviour in crises. The cross-sectional version lacks it.

7.5 The design space

The diagram lays out the four slots and their main choices.



The two teal boxes are where the return comes from. The two grey boxes are where the literature concentrates.

§7 Key takeaways

1. Expand every rule into its **return-space kernel**. It takes a few lines of code, and it reduces many named indicators to one comparable object.
2. **The entries in the kernel table are exact identities**, not analogies. That includes the equality between the moving-average rule's expected P&L and half the excess of the variance ratio over one.
3. A crossover's advantage is that it **down-weights the most recent return**. Describing it as "fast versus slow" omits the only property that matters.
4. Check every kernel for **unintended negative lobes**. Some popular multi-average rules contain hidden mean reversion.
5. Compare rules by the **centre of mass** of their kernels, not by their parameter names. SMA(200) and EWMA(span 200) differ by 50% in effective horizon.
6. **The ordering of effort is volatility scaling, then correlation structure, then timescale, then response, then kernel shape.** This inverts the attention these receive in practitioner writing.
7. Cross-sectional and time-series momentum share a name and are different strategies. The net directional exposure of the time-series version is the source of most of its distinctive behaviour.

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8. From signal to portfolio

8.1 Effort ordering

The ordering comes before any detail. The most common way to waste six months on a trend system is to spend them on the indicator.

1. **Data correctness.** Correct data is a precondition, not a component. A single mis-stitched futures roll can create or destroy an entire market's apparent edge (§8.2). Data correctness deserves more effort than everything below it combined.
2. **Volatility estimation and scaling** (§8.3). This is rank 1 of §7.3.
3. **Universe construction** (§8.6): which markets, and how correlated they are.
4. **Portfolio risk aggregation** (§8.6).
5. **Kernel timescale** (§5.8).
6. **Cost model and trading rate** (§8.7, §8.8).
7. **Response function** (§8.4).
8. **Kernel shape.** Last.

A mediocre indicator on clean data with correct sizing across 60 markets beats a superb indicator on dirty data across five.

8.2 The data layer

Trend-following trades futures. Futures data have three problems that equity data do not have.

Contract stitching. A futures "price series" is a constructed sequence of distinct contracts joined together. At each roll, the front and next contracts trade at different prices, and the gap is not a return. Four conventions exist, and the first is always a mistake:

Method	What it does	Use for
Unadjusted	Concatenate raw prices	Nothing. It injects a fake return at every roll

Method	What it does	Use for
Back-adjusted (panama)	Shift the older history by the cumulative roll gaps	Rules in price *differences*. Absolute changes survive; percentage and log returns do not, and *levels* are meaningless and can go negative
Ratio-adjusted	Multiply the older history by the cumulative roll ratios	Signal generation. Log returns survive exactly and levels stay positive
Return-stitched	Compute returns within contract, concatenate returns	The cleanest. Reconstruct a level series if a rule needs one

[Practice] Recommendation: use return-stitching, and rebuild a synthetic level series from the returns. The result equals the ratio-adjusted series up to a constant factor. The method makes the handling of the roll explicit instead of implicit. It also removes the temptation to use price *levels* in a rule, because back-adjusted series silently corrupt levels. Back-adjusted data also cause a specific error. A breakout rule computed on a back-adjusted series compares today's price with a historical extreme that the accumulated roll gaps have shifted. Market participants did not see that extreme.

Roll timing and look-ahead. Rolling on a fixed calendar day is reproducible. Rolling at the crossover of open interest is realistic, but it requires care. Publishers often release open-interest data with a lag, and using the data on the same day is look-ahead. **[Practice]** Roll on a fixed schedule, a few days before first notice, and lag any decision based on volume or open interest by at least one bar.

Non-synchronous closes. In a global futures panel, markets close across 18 hours. Correlations computed on closes labelled with the same date are biased downward for markets in distant time zones. And a signal computed at one market's close may use another market's stale price. **[Practice]** Either sample all markets at a common wall-clock time, or accept the bias and estimate correlations from overlapping multi-day returns, which reduces it.

Survivorship. The sample must include delisted contracts, failed currencies, and markets that stopped trading. A universe of markets that still trade in 2026 is biased toward survivors. Survivors and non-survivors differ most in trend-following's tail.

8.3 Volatility estimation

Volatility estimation is the most important component. The task is to forecast the next bar's return volatility from information available now.

The standard choice is an EWMA of squared returns, $\hat{\sigma}_t^2 = (1-\alpha_v)\hat{\sigma}_{t-1}^2 + \alpha_v r_t^2$, with the decay α_v corresponding to a span of roughly 20–60 bars. RiskMetrics writes the same recursion with the persistence $1 - \alpha_v$ in front and calls that constant λ. Its conventional value, 0.94, is a span of about 32 bars. This is the same estimator, written with the complementary constant. Three choices need deliberate attention:

- **Speed.** A fast estimator tracks changes of regime. It also adds its own noise to the position and therefore to turnover. A slow estimator is stable. It leaves positions oversized going into a spike in volatility. **[Practice]** Blend the two with $\hat{\sigma}_t = \frac{1}{2}\hat{\sigma}_t^{\text{fast}} + \frac{1}{2}\hat{\sigma}_t^{\text{slow}}$. Use a fast leg with a span of about a month and a slow leg with a span of six months. Average the volatilities, not the variances. Carver recommends this blend. It is meaningfully better than either estimator alone, because the slow component stabilises the estimate during transient spikes.
- **Range estimators.** The Parkinson and Garman–Klass estimators use the bar's high and low. They are several times more efficient than close-to-close estimators for the same window. Baltas and Kosowski (2013) find that more efficient volatility estimation cuts portfolio turnover by more than a third, without a significant loss of performance. Given §8.8, that is a large effect. These estimators are sensitive to bad prints in the high and low, so winsorise them.
- **The floor.** In a quiet market, a volatility estimate can become arbitrarily small, and $\sigma^\star/\hat{\sigma}_t$ then becomes arbitrarily large. **Always floor the estimate** at something like the 10th percentile of its own *trailing* history, and cap the resulting position independently.

Do not compute the volatility scaling with full-sample data. This is the most common look-ahead in trend backtests. It is very hard to see, because the resulting equity curve looks plausible rather than absurd. Every $\hat{\sigma}_t$ must use only $\{r_s : s \le t\}$.

The table shows how much volatility scaling matters. It comes from a simulation, run for this chapter, with realistic volatility clustering. The signal is identical in every row.

Position rule	Sharpe
Raw signal x_t, no scaling	0.04
$x_t/\hat{\sigma}_t$	0.21

Position rule	Sharpe
$\text{sign}(x_t)$	0.12
$\text{sign}(x_t)/\hat{\sigma}_t$	**0.36**

$\text{sign}(x_t)/\hat{\sigma}_t$ beat $x_t/\hat{\sigma}_t$ for the following reason. Volatility contaminates the *magnitude* of the raw signal. A big x_t may mean a strong trend or merely a volatile market. So dividing a slow trend estimate by a fast volatility estimate leaves noise in the ratio. Taking the sign separates the two effects. **The direction comes from the slow filter, and the size comes from the volatility estimate.** A response sensitive to magnitude is still desirable for the convexity (§1.5). To get one, first normalise the signal by its *own* running scale. Then apply the response. Then scale by volatility.

8.4 The response function

g maps the normalised signal $z_t = s_t/\hat{\sigma}_t^{(s)}$ to an exposure. Here $\hat{\sigma}_t^{(s)}$ is the running standard deviation of the raw signal *itself*, estimated over a long window. It is the signal's own scale, a different object from the return volatility $\hat{\sigma}_t$ of §8.3. Dividing by it makes z_t roughly unit-variance, so that one g and one set of caps mean the same thing in every market. The table lists four canonical shapes.

$g(z)$	Convexity	Turnover	Notes
z (linear)	Highest	Highest	Unbounded position on a signal outlier
$\text{clip}(z, -c, c)$	High	High	The standard compromise, with $c \approx 2$
$\text{sign}(z)$	None	Lowest	Discards magnitude, and with it the skew
$z\,e^{-z^2/4}/0.89$	Moderate	Lowest of the smooth ones	Fades extreme signals back toward zero

The last row is the response function popularised by Baz and co-authors (2015), and used in the deep-learning trend literature. It peaks at $z = \sqrt{2}$, where $ze^{-z^2/4} = \sqrt{2}\,e^{-1/2} \approx 0.86$. So their constant 0.89 puts the maximum exposure just under 1. Beyond $|z| = \sqrt{2}$, the response decays toward zero. This decay is the fading. In simulations run for this chapter, it achieved a marginally higher Sharpe ratio than the capped-linear response, with roughly 30% lower turnover. After costs, that is a real improvement.

Whether *fading* extreme signals is right is an empirical question. Lempérière and co-authors (2014) find a clear **saturation** effect in the data. The return to a trend signal grows less than linearly and flattens for large signals. They interpret this as fundamentalist traders entering only once the mispricing is large enough to be worth trading against. The finding supports capping or a bump shape, and it argues against pure linearity. **[Contested]** The finding comes from a single study by an interested party. The mechanism is plausible, and the capping convention is nearly universal in practice regardless.

The response function also determines the **hit rate** and the **skew** (§5.7). Choose them deliberately instead of discovering them afterwards. A fund whose investors will redeem after two losing years in three should probably not run a pure linear response, whatever its Sharpe ratio.

8.5 Position sizing

The components combine into a position size. For market i, with signal $z_{i,t}$, price $P_{i,t}$ and contract point value V_i, the number of contracts is

$$n_{i,t} \;=\; \frac{\text{Capital} \times \sigma^{\star}_{\text{portfolio}} \times \text{IDM} \times v_i \times g(z_{i,t})}{\hat{\sigma}_{i,t}\sqrt{A}\; P_{i,t}\, V_i\, \text{FX}_i}$$

The terms are as follows:

- $\sigma^{\star}_{\text{portfolio}}$ is the *annualised* risk target;
- v_i is market i's weight in the portfolio, and the v_i sum to one;

- IDM is the *instrument diversification multiplier* of §8.6;
- $\widehat{\sigma}_{i,t}\sqrt{A}$ annualises the per-bar volatility estimate to match the target;
- FX_i converts the contract's currency to the book's.

Everything below the line is the annualised risk of one contract, expressed in the book's currency, so the quotient is a contract count. **Every term except $g(z_{i,t})$ concerns risk and not the view.**

Two limits belong here, and users routinely omit them:

- **A position cap per market**, independent of the volatility scaling, so that a volatility estimate near zero cannot produce an absurd position.
- **A liquidity cap**, as a fraction of average daily volume or open interest. It binds in exactly the markets that trend best.

8.6 Portfolio construction

The Sharpe multiplier of §5.9 is the source of the return. So portfolio construction is rank 2 in the ordering of effort.

The diversification multiplier. Give each of N markets an equal $1/N$ share of the risk budget, so that their standalone volatilities sum to $\sigma^\star$. Suppose their strategy returns have average pairwise correlation $\bar{\rho}$. The positions then partly cancel. The portfolio's volatility is $\sigma^\star\sqrt{(1+(N-1)\bar{\rho})/N}$, which is much lower than $\sigma^\star$. Reaching the target requires scaling up by

$$\text{IDM} \;=\; \sqrt{\frac{N}{1+(N-1)\bar{\rho}}}\,,$$

This is the same factor as the Sharpe multiplier. For 50 markets at $\bar{\rho}=0.15$ it is 2.45. The book must run 2.45 times the naive gross exposure to achieve its target risk. An error in either direction is large. If the IDM is too low, the book runs at half its intended risk for years. If it is too high, a spike in correlation delivers double the intended risk.

The danger is that $\bar{\rho}$ is unstable. Trend systems converge to the same positions when many markets trend together. A shock then hits several of those positions at once. **[Fact]** Realised correlations rise sharply in stress, both between trend-following programs and between one program's positions across markets. An IDM computed from a calm-period correlation therefore overestimates exactly when overestimating is dangerous.

The three defences below are listed in increasing order of intrusiveness:

1. **Estimate $\bar{\rho}$ on a long window, and cap the IDM**, commonly at 2.5. This is crude and effective.
2. **Use signed correlations.** This refinement comes from Baltas and Kosowski (2013). The alignment of the *positions* matters, and the correlation of the markets alone does not. Two negatively correlated markets held with opposite signs are a concentrated position, not a diversified one. Compute the correlation of $\text{sign}(z_i)r_i$ with $\text{sign}(z_j)r_j$.
3. **Optimise directly on a shrunk covariance matrix**, with a Ledoit–Wolf or factor-model estimator. This is better in principle. In practice, the estimation error at $N \approx 50$–100 with a few years of data usually exceeds the gain. The resulting weights are also unstable enough to add turnover.

[Practice] Recommendation: use (1) and (2), and skip (3) unless N is large and the history is long.

Sector caps. Impose structural limits beyond correlations. Allow no more than some fraction of risk in bonds, in energy, or in a single currency bloc. These limits are crude. They protect against a wrong correlation estimate, which is the costly case.

Market selection. By §5.9, reducing $\bar{\rho}$ from 0.15 to 0.05 is worth more than trebling N. In practice, prefer adding a market from an under-represented sector to adding a second contract in a well-covered one. Be sceptical of universes whose apparent breadth consists of many correlated instruments.

8.7 Trading rate, buffering, and the optimal-execution result

A naive implementation trades to its exact target every bar. That generates enormous and mostly pointless turnover, because most of the change in the target is the volatility estimate's own noise.

Gârleanu and Pedersen (2013) give the optimal policy. With quadratic transaction costs and mean-reverting predictors, the policy is:

> **Trade partially toward an "aim" portfolio, where the aim over-weights the more persistent signals.**

$$\pi_t \;=\; (1-\kappa)\,\pi_{t-1} \;+\; \kappa\,\text{aim}_t$$

The trading rate $\kappa \in (0, 1]$ increases with risk aversion and decreases with transaction costs. Trade faster when risk matters more, and slower when trading costs more. Two consequences follow:

1. **Never rebalance fully to the target.** The optimal policy is always a partial adjustment.
2. **Signals with different decay rates deserve different weights in the aim than in the unconstrained optimum.** A fast signal loses its edge before the fund finishes building the position. Down-weight such a signal *in the target*, and do not just trade it slowly.

[Practice] The practical approximation used in the industry is a **no-trade buffer**. Compute the target position, and trade only if it differs from the current position by more than some threshold. Common thresholds are 10% of the average position, or half the position's own volatility. This captures most of the benefit of the optimal policy, without the need to calibrate a model. In simulations run for this chapter, a buffer of this size cut turnover by 30–50% at a cost to the Sharpe ratio in the low single digits of percent.

8.8 Costs and capacity

Costs are a binding constraint on the design of trend-following, not a minor deduction from its return. The table comes from a simulation, run for this chapter, that sweeps the kernel timescale and applies a proportional round-trip cost.

EWMA span	Turnover	Gross SR	Net SR @ 1bp	@ 5bp	@ 20bp
20	13.5×/yr	0.20	0.16	-0.02	-0.68
80	7.1×/yr	0.32	0.29	0.20	-0.14
320	3.9×/yr	0.37	0.36	0.31	0.12

Costs make fast trend rules unprofitable, and slow rules remain profitable. A round trip of 5 basis points is realistic for liquid futures, including market impact. At that cost, a rule with span 20 is already unprofitable, while a rule with span 320 keeps 84% of its gross Sharpe ratio. This explains why production systems run lookbacks of 1–12 months. The explanation is better than the claim that "that is where the trends are".

Three components need modelling:

- **Spread and commission.** These are known, and small in liquid futures, often under a basis point. They are the easy part.
- **Market impact.** It scales as roughly the square root of participation (Tóth et al., 2011). Trading a quantity Q against an average daily volume ADV costs, per unit traded, on the order of $\sigma\sqrt{Q/\text{ADV}}$. This cost is *not* linear, so doubling the assets under management more than doubles the cost.
- **Slippage against the signal.** Trend followers trade in the direction the market has just moved. So they systematically take liquidity in the direction of price pressure. Cost estimates from symmetric samples underestimate what a trend program pays.

On capacity. By the square-root law, capacity is finite, and the constraint binds at the level of the portfolio. Gross profit grows linearly in size. Total impact cost grows like $\text{AUM}^{3/2}$, because the quantity traded is proportional to AUM and the cost of each unit grows like $\sqrt{\text{AUM}}$. So net profit behaves like $a\,\text{AUM} - c\,\text{AUM}^{3/2}$. It is concave, has a maximum, and falls beyond it. **[Contested]** Frazzini, Israel and Moskowitz (2018) use roughly 1.7 trillion dollars of live executions. They find realised costs an order of magnitude below the academic proxies. This would move the capacity constraint to much larger sizes. Their firm runs these strategies at scale. That is why the data exist, and it is also why the result deserves caution.

8.9 A reference implementation

The listing gives the whole pipeline, in the order it must execute.

```
for each bar t:
    # --- 1. data, all using information available at t ---
    returns[i]    = stitched log return of market i          # §8.2
    sigma[i]      = blend(ewma_vol(returns[i], span=32),
                          ewma_vol(returns[i], span=180))    # §8.3
    sigma[i]      = max(sigma[i], vol_floor[i])

    # --- 2. signal ---
    fast[i]       = ewma(price[i], span=S)                   # §6.3
    slow[i]       = ewma(price[i], span=4*S)
    raw[i]        = (fast[i] - slow[i]) / (sigma[i] * price[i])
    z[i]          = raw[i] / ewma_sd(raw[i], span=2500)      # unit-scale signal
    g[i]          = clip(z[i], -2, +2)                       # §8.4

    # --- 3. sizing ---
    target[i]     = capital * vol_target * IDM * weight[i] * g[i]
                    / (sigma[i] * sqrt(A) * price[i] * point_value[i] * fx[i])
    target[i]     = clip(target[i], -pos_cap[i], +pos_cap[i]) # §8.5
    target[i]     = clip(target[i], -adv_cap[i], +adv_cap[i]) # liquidity

    # --- 4. portfolio overlay ---
    scale         = min(1, max_gross / sum(|target|))
    scale        *= drawdown_multiplier(equity_curve)         # §11.4
    target        = target * scale

    # --- 5. trade ---
    if |target[i] - position[i]| > buffer[i]:                 # §8.7
        trade to target[i]
```

Four features of this listing matter, because users often get them wrong:

- Every quantity uses only data available at t.
- The volatility estimate appears **twice**. It normalises the signal once and sizes the position once. The two uses differ in concept. The first puts `raw` into units of daily sigma. The second turns a risk target into a contract count.
- The position caps are independent of the volatility scaling, so a broken $\widehat{\sigma}$ cannot produce an unbounded position.
- The trade decision is a buffered comparison, not an assignment.

Set the signal parameters from §5.8 and not from a search. Use $S \approx \tau/2$, where τ is the estimate of trend persistence in bars, and a slow span of about $4S$.

§8 Key takeaways

1. **The ordering of effort is data, volatility, universe, aggregation, timescale, costs, response, kernel shape.** A mediocre indicator on clean data across 60 markets beats a superb one on dirty data across five.
2. A futures price series is a construction, not an observation. Use return-stitching, roll on a schedule, and never let a rule see a back-adjusted *level*.
3. Blend a fast and a slow volatility estimate, floor it, and cap the position independently of it. Range estimators cut turnover by a third without a significant loss of performance.
4. Compute every $\widehat{\sigma}_t$ causally. Full-sample volatility scaling is the most common invisible look-ahead in this literature.
5. **Take the direction from the slow filter and the size from the volatility estimate.** Dividing a slow signal's magnitude by a fast volatility estimate leaves noise in the ratio.
6. The diversification multiplier $\sqrt{N/(1+(N-1)\bar{\rho})}$ is both the source of the Sharpe ratio and the largest single risk in sizing, because $\bar{\rho}$ rises in stress. Cap it, and use signed correlations.
7. **Never rebalance fully.** Partial adjustment toward an aim portfolio is optimal under costs, and a no-trade buffer captures most of the benefit without calibration.
8. **Costs determine the horizon.** At a 5bp round trip, a rule with span 20 is unprofitable, and a rule with span 320 keeps 84% of its gross Sharpe ratio. This, not the location of "the trends", is why production systems are slow.

9. When trends fade and fail

9.1 Three different claims

"Trend-following stops working" conflates three claims. They have different evidence, different timescales and different remedies, as the table shows.

Claim	Timescale	Remedy
Within-trade decay — this particular trend is exhausting	Days to weeks	Exit and sizing rules; nothing structural
Regime dependence — trends are absent in this environment	Months to years	Risk reduction, diversification, patience
Secular decay — the premium itself has shrunk	Decades	Change the strategy or the business

Confusing the first claim with the third causes a classic failure. Investors abandon a strategy after a normal drawdown. Confusing the third with the first causes the opposite failure. Investors keep allocating patiently to a strategy that no longer works.

9.2 Within-trade: the sign of predictability flips with horizon

The first row of the table in §9.1, *this trend is exhausting*, usually appears as a claim about one position. It rests on a claim about horizons. The quantity a trend rule is long, $\sum_{k\geq 0} w_k \gamma_{k+1}$ from §5.2, has no single sign. The lags the kernel touches decide whether that inner product is positive or negative. This holds before any question of estimation arises.

[Fact] Return autocorrelation is a function of lag. It has no single value, and its sign changes. The table gives the broad pattern. The pattern varies substantially by asset class.

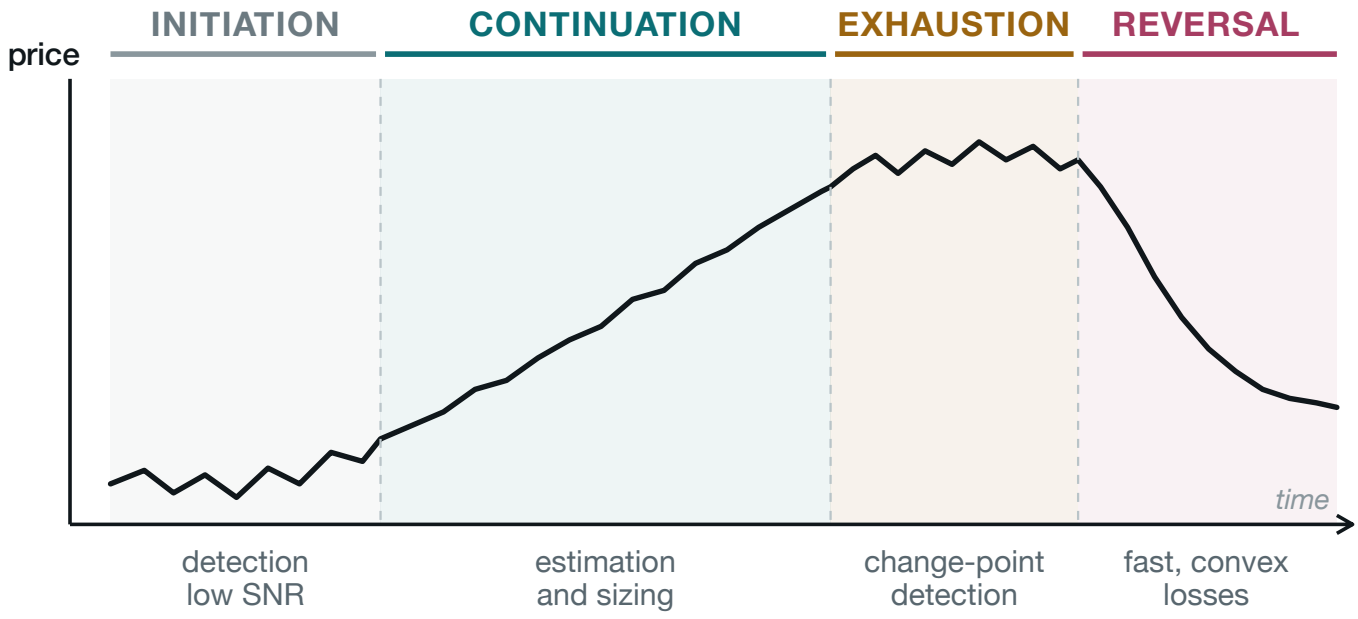
Horizon	Dominant sign	Mechanism	Consequence for a trend rule
Seconds to minutes	Positive in order flow, near zero in price	Order splitting	Not accessible to a daily system
1–5 days	**Negative** (reversal)	Liquidity provision, bid-ask bounce	Fast rules trade against this, and they also pay the costs of §8.8
1 month	Negative in cross-section	Short-term reversal	The reason equity momentum skips the last month
2–12 months	**Positive**	Underreaction, flow, macro persistence	Where trend-following earns its return
1–3 years	Weak	—	Nothing to harvest
3–5 years	**Negative** (reversal)	Overreaction correction, valuation	Where a slow trend rule loses money

This is the *structural* reason a trend rule has an optimal horizon. It is stronger than the estimation argument of §5.8. Even with perfect estimation, a rule whose kernel has mass at lags of three days puts positive weight on a negative autocorrelation. The argument adds to the cost argument of §8.8 and does not replace it. Costs bind in production. But costs alone would not show that the fast end of the kernel has the wrong sign. They would show only that it is expensive.

The practical corollary, from §7.2, is to **check where the kernel's mass actually is.** The weight in the reversal zone depends on the kernel's *shape*, not on the window in its name. Two rules with the same nominal horizon can differ by an order of magnitude. Price minus a 200-day SMA has the descending-ramp kernel $w_j = (n-1-j)^+/n$ of §7.2, which peaks at lag 0. It puts 4.9% of its total weight at lags 1–5, on a negative autocorrelation. A crossover matched to the same effective horizon, SMA(50) $-$ SMA(148), has a centre of mass of 65 bars against the ramp's 66. It peaks at lag 49 and puts only 0.5% of its weight at lags 1–5. That is a ninth of the exposure to the wrong sign, with no loss of horizon. This explains a large part of why band-pass crossovers beat price-versus-average rules net of costs.

9.3 The trend life cycle

[Practice / Hypothesis] The four-phase description below organises the discussion and has partial empirical support. It is not an established taxonomy. It identifies *which statistical problem* each stage poses. The figure and table set out the phases.



Phase	Statistical problem	Signal quality	What to do
Initiation	Detection: is this the onset of drift or noise?	Weak, and most detections are false	Small size, wide tolerance
Continuation	Estimation and sizing	Strongest	Full size, volatility-target
Exhaustion	Change-point detection, the hardest of the four	Strong but decaying	Reduce, tighten
Reversal	—	Inverts abruptly	Be flat or reversed

Initiation. The signal-to-noise ratio is worst here by construction, because the new state has the fewest observations. The lookback sets the balance between Type I and Type II errors. No parameter removes the trade-off. Only the cost structure says which point on the trade-off curve to choose (§8.8). A short lookback detects real onsets sooner, and it also detects false onsets constantly. A long lookback does the reverse. Most detections will be false. The trade hit rate of 25% in §5.7 reflects this, because it counts the losing trades.

Exhaustion. Candidate markers **[Practice, weakly supported]** are:

- rising volatility with flat absolute price progress, which is a falling efficiency ratio (§6.9)
- negative acceleration while the trend estimate is still positive
- divergence between price extremes and oscillator extremes
- crowding measures.

None of these is a robust standalone signal in the published evidence. **Treat exhaustion detection as a reason to reduce risk, never as a reason to take the opposite position.** The evidence does not support the reversal trade, and the payoff structure is unfavourable to it.

9.4 Secular decay: whether the premium has shrunk

The question is open, and both sides get a full statement below.

The case that it has decayed.

- Realised CTA index returns were poor from roughly 2009 to 2019. A decade is long enough that calling it a "normal drawdown" is hard to believe.
- Assets in the strategy grew by an order of magnitude between 2000 and 2015. If the premium compensates for providing liquidity to slow flows, more capital in the strategy should compress it.
- Researchers published the mechanism. Moskowitz, Ooi and Pedersen (2012) and its successors made the signal explicit. Cheap products followed, and fees collapsed. Other anomalies have shown the same pattern after publication.
- **[Hypothesis]** Central-bank intervention from 2009 suppressed the volatility and the dispersion of rates on which trend-following in fixed income depends.

The case that it has not.

- The 2009–2019 window is short relative to the statistical resolution of §5.9. Twenty years cannot distinguish a Sharpe ratio of 0.4 from 0, and a decade certainly cannot.
- 2022 was one of the best years in the strategy's history, driven by trends in bonds and energy. A decayed premium does not produce such a year.
- The long-history studies (§4.2) find the effect across eras with very different levels of participation, including eras with essentially no systematic capital at all.
- Arguably the most crowded trade of 2009–2019 was the short-volatility complex, not trend-following. Volatility stayed suppressed and reverted quickly to its mean. That environment hurt trend and favoured the opposite of trend.

The view taken here. The premium is probably smaller than the sample before 2000 suggests, and probably not zero. Two specific components have almost certainly decayed. One is fast trend. Competition through execution technology eliminated its excess return, and it now falls inside the cost constraint of §8.8. The other is the return investors receive after fees, which competition compressed. That is good for investors and bad for managers. People often mistake it for decay of the underlying premium.

9.5 Crowding and capacity

[Contested] Crowding is easier to assert than to measure. Three observable proxies exist:

1. **Positioning data.** Commitments-of-Traders reports classify the positions of managed money. They are crude, weekly and heavily lagged, but directionally informative.

2. **Return correlation among trend programs.** Rising correlation of CTA returns with each other, and with a generic trend replicator, is evidence that the programs converge on the same positions.
3. **Forced unwinds.** Crowding appears as *conditional* behaviour. Positions that a generic trend model would hold reverse sharply, at moments when the strategy is losing.

Crowding **changes the shape of the loss distribution**. It also lowers the average return, mildly, through impact. But the loss distribution is the main concern. Many participants holding the same positions, with similar risk limits, will deleverage at the same time. That turns a moderate adverse move into a large one. This is tail risk, so §11.7 treats crowding as a topic in risk management rather than in return forecasting.

9.6 What would count as evidence

The nearest anecdote settles this question too easily. This section states in advance what evidence would change the view.

Evidence that the premium is gone:

- The variance-ratio profile of a broad futures panel converging to 1 at all horizons over a rolling 10-year window, tested on demeaned returns. This is the direct measurement, and by §5.4 it *is* the expected P&L.
- A statistically significant negative trend in the per-decade Sharpe ratio of trend-following across the full sample from 1880 to the present, robust to the choice of universe.
- Trend performance failing to recover in a period with large, sustained macroeconomic dispersion. 2022 was the natural test, and trend did not fail it.

Evidence that it persists:

- Continued positive excess in the variance ratio at horizons of 2–12 months, on demeaned returns, in markets outside the original sample.
- Continued performance in markets with low systematic participation, where the crowding explanation does not apply.

[Practice] Recommendation: run the variance-ratio monitor as standing infrastructure. It is cheap. Compute $\mathrm{VR}(q)$ on demeaned returns for q from 5 to 250 bars, per market and pooled, on a rolling window. It measures directly the source of the strategy's return. It signals a change long before the P&L does, and it requires no backtest.

§9 Key takeaways

1. Distinguish within-trade decay, regime dependence and secular decay. They have different timescales and opposite remedies. Conflating them causes the classic error of capitulating at the bottom.
2. **Autocorrelation changes sign with horizon**: negative at days, positive at 2–12 months, and negative again at 3–5 years. A kernel with mass in the wrong zone loses expected P&L on that mass.
3. This is a stronger argument for the standard horizon of 1–12 months than the estimation argument, and it reinforces the case for band-pass kernels.
4. Detecting exhaustion is the hardest of the four life-cycle problems, and no marker for it is well supported. **Use it to reduce risk, never to reverse.**
5. Secular decay is an open question. Fast trend has decayed, and the return investors receive after fees has compressed. The underlying premium probably persists at a reduced level.
6. A decade of poor performance is statistically uninformative at this Sharpe ratio. Ten years cannot distinguish 0.4 from 0.
7. Crowding mainly affects the **shape of the loss distribution**, not the mean. Simultaneous deleveraging turns moderate moves into large ones.
8. **Run a rolling variance-ratio monitor on demeaned returns.** By §5.4, it measures the expected P&L directly, and it reacts long before the equity curve does.

---

10. Evaluation: testing a trend system rigorously

10.1 The staged tests

Test in stages, cheapest first, and stop at the first failure. Each stage answers a different question. Running them out of order wastes effort on systems that were never viable. The table lists the stages.

Stage	Question	Test	Kill criterion
0	Is there anything to harvest?	VR(q) profile on demeaned returns, pooled	No excess at any q
1	Does the signal carry information?	Correlation of z_t with r_{t+1}, pooled across markets	Pooled correlation indistinguishable from 0
2	Does it survive the drift confound?	Repeat stage 1 on demeaned returns; alpha vs constant-long	Effect disappears
3	Does it survive as a portfolio?	Full backtest with sizing and aggregation	Sharpe below a constant-long position at the same average exposure (§5.3)
4	Does it survive costs?	Re-run at $1\times$, $3\times$, $10\times$ the cost estimate	Sign flips within $3\times$
5	Does it survive the search?	Deflated Sharpe; walk-forward on untouched data	Deflated Sharpe below 0.95 (§10.5: it is a probability)
6	Does it survive implementation?	Paper trade with real fills, real rolls, real timing	Live/backtest gap unexplained

Stage 0 is unusual, and it uses the result of §5.4. The expected P&L of the moving-average rule *is* half the excess of the variance ratio over one. So measuring the variance ratio shows the opportunity before any backtest exists. If the pooled VR profile on demeaned returns is flat, the data hold nothing to find, and no rule will find it.

The kill criterion of stage 4 is deliberately harsh. Trend P&L is a small residual between two large terms (§5.5). Cost estimates are normally off by a factor of two, and that error moves the answer materially. A system that works only at the central cost estimate does not work.

10.2 Errors specific to trend

Trend systems face the following hazards in addition to the usual hazards of backtesting.

The drift confound (§5.3). Demean the returns. This is the largest hazard.

Full-sample volatility scaling. A $\widehat{\sigma}_t$ computed on the whole sample puts future information into every position. The resulting curve looks *plausible*, so this error survives review.

Inference on overlapping windows. A 12-month signal sampled daily produces observations that overlap by 251 days. Ordinary standard errors on such a series are too small by roughly $\sqrt{L}$, a factor of 16 for $L = 252$. Use Newey–West with a lag at least as long as the overlap, or sample without overlap, or use a block bootstrap.

Survivorship in the universe. Include markets that stopped trading.

Look-ahead in rolls and timing. Three examples occur. A system uses the settlement price to decide a trade executed at that settlement. It rolls on the same day's open interest. It uses a close that occurs, in another time zone, after the decision time.

Backtesting the smoother, not the filter. This is the look-ahead error of §6.7.

Symmetric cost models. Trend systems buy after the market has risen, so they trade with the flow. Costs estimated from a symmetric sample are optimistic.

Reporting trades rather than time. Trade-level statistics (§5.7) look informative, but the definition of a trade is arbitrary. It can mean position flips, sign changes or round turns. Report returns indexed by time, and use trade statistics only for diagnostics.

10.3 Synthetic paths, where the answer is known

Synthetic paths are the most underused validation tool. The local level model of §5.8 has a known trend structure, so it generates paths with a *specified* τ and θ. A researcher can then check the implementation against the known optimum.

A synthetic test catches three things that a backtest cannot:

- **Implementation bugs that look like alpha.** A system that beats the theoretical optimum on synthetic data with a known answer has look-ahead. This is the most effective available test for look-ahead, because it supplies a hard ceiling for comparison.
- **Whether the estimator is even capable** of extracting a known signal at the available data length. If it cannot find a trend that is *known* to be there, no amount of testing on real data will help.
- **Sizing errors.** Feed the system a path with known volatility, and check that the realised risk matches the target.

Generate paths at the believed (τ, θ), run the full pipeline, and compare the result with the Kalman-optimal benchmark. A well-built system should reach 80–95% of the benchmark. Substantially more means a bug. Substantially less means the kernel or the sizing is wrong.

10.4 Bootstrapping a path-dependent strategy

The standard IID bootstrap is useless here. It destroys exactly the serial dependence the strategy exists to harvest, so it always finds that the strategy has no edge.

Two constructions work:

- **The stationary block bootstrap** (Politis and Romano), with a mean block length comfortably longer than the kernel's centre of mass. Resampling blocks preserves the dependence within each block. Use it to get honest confidence intervals on the Sharpe ratio.
- **The null bootstrap.** Deliberately destroy serial dependence by shuffling returns within each market, and rerun the *whole* pipeline. This gives the distribution of the strategy's performance under the null of no predictability. The shuffle leaves the drift and the selection procedure intact. Shuffling blocks of standardised returns and re-imposing the volatility path also preserves the volatility clustering. **Compare the reported Sharpe ratio with this distribution, not with zero.** For a linear-response rule, the distribution is right-skewed (§1.5). Its median lies *below* zero, and its right tail extends far above zero. People notice the median first, and it makes the null look easy to beat. The tail matters more. A Sharpe ratio that looks comfortably positive against zero can still be an unremarkable draw from a strategy with no edge at all.

10.5 Multiple testing

A typical research process tests many strategies. It tests n_{span} span choices, n_{resp} response functions, n_{vol} volatility estimators and n_{univ} universes. The research compares the *combinations*, so the effective trial count is the product. Four of each gives 256. The maximum Sharpe ratio over that many trials is biased upward, even when every trial is worthless.

The **deflated Sharpe ratio** of Bailey and López de Prado is the standard correction. It computes the Sharpe ratio that the *best* of that many trials would be expected to reach under the null of no skill. This benchmark rises with the trial count. The method adjusts for the skew and kurtosis of the returns. It reports the probability that the observed Sharpe ratio beats the benchmark. The output is a probability, so the required value is near 1, not merely above 0. Harvey and Liu argue, on the same grounds, for reducing reported t-statistics substantially.

Two points are specific to trend research.

The trial count includes history. The industry selected trend rules over 50 years, on largely the same price series. The independent trials include those that a given researcher ran and everyone else's trials as well. This is the strongest form of the Sullivan–Timmermann–White critique. No statistical fix exists. Only out-of-sample data help, meaning new markets and new eras.

Flat optima reduce the damage. The flatness result of §5.8 helps here. If performance is insensitive to a parameter, the maximum over a sweep of that parameter is only slightly above the average. So the selection bias from the sweep is small. [Practice] **Recommendation: report the average over the sweep, not the maximum.** For a flat objective, that costs almost nothing and removes most of the bias. It is a rare case where the honest choice costs almost nothing.

§10 Key takeaways

1. **Stage 0 is a variance-ratio profile, not a backtest.** By §5.4, it measures the opportunity directly. A flat profile means no rule will work.
2. Demean every market's returns before any test. Undemeaned trend results are uninformative.
3. Overlapping windows inflate t-statistics by roughly $\sqrt{L}$. Correct for this, or resample.
4. **Test on synthetic paths with a known answer.** Beating the Kalman-optimal benchmark proves look-ahead, and it is the best such detector available.

5. Compare the Sharpe ratio with a **null bootstrap that preserves the drift and volatility structure**, not with zero. For linear-response rules, the null is right-skewed.
6. Stress-test costs at 3× and 10×. A system that works only at the central estimate does not work.
7. **Report the average across the parameter sweep, not the maximum.** With a flat optimum, this costs almost nothing and removes most of the selection bias.
8. The effective trial count includes 50 years of industry search over the same data. Only genuinely new markets and eras fix that.

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11. Pitfalls and risk management

11.1 The pitfall catalogue

The catalogue groups the pitfalls by where they cause damage. The ones at the research stage cost time. The ones in implementation cost money. The ones in live trading cost the business.

#	Pitfall	Why it happens	Detection
Research stage			
1	Drift confound	The test statistic is biased positive under the null (§5.3)	Demean; compare to constant-long
2	Full-sample volatility scaling	Looks like preprocessing but is look-ahead	Recompute causally; the curve should get worse
3	Backtesting a two-sided smoother	HP and ℓ_1 fits use future data (§6.7)	Re-solve on expanding windows
4	Overlapping-window t-statistics	Inflated by $\sqrt{L}$	Newey–West or block bootstrap
5	Universe survivorship	Dead markets silently excluded	Count markets per year; it should not only rise
6	Optimising the lookback	A flat, noisy objective with many trials (§5.8)	Report the sweep average
7	Hidden negative kernel lobes	Invisible in price space (§7.2)	Plot the return weights
Implementation			
8	Roll gaps as returns	Unadjusted series inject fake returns	Check return distribution near roll dates
9	Back-adjusted levels in a level rule	Breakout levels shifted by roll accumulation	Never use adjusted levels; use returns
10	Volatility estimate collapse	Quiet market $\Rightarrow$ divide by near-zero	Floor $\widehat{\sigma}$; cap positions separately
11	IDM from calm-period correlations	$\bar{\rho}$ rises in stress (§8.6)	Cap the IDM; use long windows
12	Symmetric cost estimates	Trend trades with the flow	Model impact asymmetrically
13	Trading to target every bar	Most of the change is estimator noise	Buffer (§8.7)
Live			
14	Gap risk in a synthetic option	The convexity requires a tradeable market (§5.6)	Stress-test with historical gaps
15	Simultaneous deleveraging	Everyone's risk limits bind at once (§9.5)	Monitor crowding; stagger limits
16	Capitulating in a normal drawdown	Ten years is statistically uninformative (§5.9)	Pre-commit to a governance rule
17	Parameter drift	Quietly re-fitting after each drawdown	Version and date every parameter change

Number 17 survives every process, so it gets its own paragraph. Re-tuning a system after each bad period fits it to the full sample, one drawdown at a time. Its live track record is therefore an in-sample result. **[Practice]** Keep a dated log of every parameter change and its justification. Compute the track record using the parameters as they stood at each point in time. The difference between that record and the backtest on current parameters directly measures how much overfitting has occurred.

11.2 Volatility targeting

Scaling positions by $\sigma^\star/\widehat{\sigma}_t$ is the core risk mechanism. The lists below state precisely what it achieves and what it does not.

What it does. [Fact]

- **It stabilises realised risk.** This is its primary purpose, and it works. In a simulation run for this chapter, volatility scaling cut the dispersion of realised P&L volatility across sub-periods by roughly three quarters.
- **It improves the Sharpe ratio when volatility is persistent and negatively related to future returns.** Harvey and co-authors (2018) document this across asset classes. The effect is strongest in equities, where the relation between volatility and return is strongest. It is much weaker in commodities and currencies.
- **It reduces the severity of the worst drawdowns**, by deleveraging into rising volatility.

What it does not do.

- **It does not help when volatility jumps rather than drifts.** The estimator uses only past data, so the position stays at full size through a discontinuous increase in volatility. Volatility targeting protects against the 2008 pattern, when volatility rose over weeks. It does not protect against the openings of February 2018 or March 2020.
- **It does not reduce risk in aggregate. It shifts risk to other periods.** Deleveraging when volatility is high means levering up when volatility is low. Fat-tailed jumps arrive from a low base in low-volatility periods.
- **It can worsen skew.** A volatility target mechanically sells into declines. Harvey and co-authors find that it improves the Sharpe ratio but has an unfavourable effect on skewness for some asset classes.

[Contested] Per §7.3, Kim, Tse and Wald (2016) argue that much of the reported time-series momentum premium comes from the volatility scaling rather than from the trend signal. If so, volatility targeting is a substantial part of the strategy rather than risk management added on afterwards. Analysts should evaluate it as such.

11.3 Stops, and a result that surprises people

Folk wisdom holds that stop losses are essential to trend-following. The theory gives a more specific answer.

Kaminski and Lo (2014) analyse when a stop-loss overlay improves the expected return of an underlying strategy. Their result:

> When returns have **positive autocorrelation**, a stop-loss rule **adds** expected return. When returns follow a random walk or mean-revert, it **subtracts** expected return.

The intuition is direct. A stop sells after a loss. If more losses tend to follow a loss, which is positive autocorrelation, selling is right. Under a random walk, the future is independent of the loss, so the stop conveys no information at all. It pays a transaction cost. Suppose the underlying has a positive drift, as in Kaminski and Lo's setting of an equity position. Then the stop also gives up that drift while the position is out of the market. Under mean reversion, a stop is worse still, because it sells exactly what is about to recover.

Two consequences matter:

1. **A stop is another trend signal, not risk management.** A stop is a fast, nonlinear trend rule, triggered by a threshold and added on top of the slow linear one. Evaluate it as a signal, with the same scrutiny, and check whether it merely duplicates the existing signal with more noise.
2. **The case for stops in trend-following is therefore self-consistent but redundant.** The positive autocorrelation that justifies the stop already justifies the underlying rule, which reduces the position anyway as the signal decays. Adding a stop mostly adds a faster kernel, with the cost profile of a fast kernel (§8.8).

None of this makes stops *dangerous*. §5.7 mentioned this asymmetry. A stop truncates the left tail, where a trend strategy has its many small losses. A profit target truncates the right tail, where the strategy earns essentially all of its profit. A stop is "redundant and mildly costly", which is a very different verdict from "ruinous". The two modifications are not symmetric.

[Practice] Stops do have a role against risks the signal cannot detect. A *hard* loss limit per position catches data errors, model failures and gap events. That is a different function from trend-following. Size it and justify it as an operational circuit-breaker, not as alpha. Set it far enough out that it never triggers in normal operation.

11.4 Drawdown control

The theoretical baseline is Grossman and Zhou (1993). They solve for the optimal investment policy under a constraint that wealth never falls below a fixed fraction of its running maximum. The solution scales exposure with the distance from the drawdown floor. As wealth approaches the floor, exposure falls toward zero.

The practical form is a multiplier on the portfolio scale, the c_t slot of the master form of §1.3 and §7.1:

$$c_t \;=\; c_0 \cdot \psi\!\left(\frac{D_t}{D_{\max}}\right), \qquad D_t \;=\; 1 - \frac{W_t}{\max_{s \le t} W_s}$$

W_t is the program's equity at t, so $D_t \in [0, 1)$ is the current drawdown from the running high-water mark. $D_{\max} \in (0, 1)$ is the drawdown chosen as tolerable. c_0 is the scale at which the program runs when there is no drawdown. The throttle ψ maps $[0, 1]$ into $[0, 1]$, with $\psi(0) = 1$, and falls as D_t rises toward $D_{\max}$. In Grossman and Zhou's solution it falls all the way to zero, though the recommendation below argues for stopping short of that. Every argument here is a ratio, so c_t is a dimensionless multiplier applied to every position at once. Three cautions apply:

- **It converts a temporary loss into a permanent one.** Deleveraging makes the drawdown permanent by removing the exposure that would recover it. Given the return distribution of §5.7, in which a handful of trades produce the entire profit, deleveraging during the recovery is expensive.
- **It interacts badly with volatility targeting.** Both deleverage in the same states, and the product of the two multipliers can take exposure to almost nothing. Model them jointly.
- **Its path dependence makes naive backtesting unreliable.** The multiplier depends on the program's own equity curve, so a small change anywhere changes everything downstream. Confidence intervals from a block bootstrap (§10.4) are essential here.

[Practice] Recommendation: use a gentle, slow-moving drawdown multiplier, with a floor of perhaps 0.5 rather than 0, engaging only well into a serious drawdown. Treat it as a tool for retaining investors and keeping the business operating, not as a way to improve returns. It does not improve returns, and the literature does not claim that it does.

11.5 Tail and gap risk

§5.6 makes this point, and this section restates it as a risk register: **a trend follower is long diffusive convexity and short gap risk.** Trading creates the convexity, and creating it requires a functioning market. The specific exposures follow:

- **Limit moves.** Commodity futures with daily limits can lock, which prevents exit for consecutive sessions. The position stays frozen at exactly the moment the model signals an exit.
- **Weekend and overnight gaps.** The largest moves in FX and rates have occurred while markets were closed. Examples are devaluations, broken pegs, referendum results and central-bank surprises. In January 2015, the Swiss franc de-peg moved roughly 30% in minutes. Positions had been sized on the prior volatility regime, and no way existed to reduce them.
- **Structural breaks in the price process itself.** In April 2020, WTI futures settled at a negative price. Every log-return pipeline in existence produced either a NaN or a nonsense number. **[Practice]** Decide in advance what the system does with a non-positive price, and test it. An untested branch here fails as an unbounded position, not as a missing data point.
- **Correlated gaps.** A single macro event gaps many markets at once, in the same direction as the positions, because the same recent trend established all of them.

Mitigations, in order of how much they actually help, are:

1. position caps that bind independently of the volatility scaling;
2. caps by sector and by factor;
3. explicit stress tests that replay historical gap events against current positions;
4. for programs large enough to afford it, purchased optionality on the largest exposures.

The last is the direct solution to the replication risk of a synthetic option. It also costs the option premium that the synthetic replication was designed to avoid.

11.6 Correlation risk

The IDM of §8.6 makes correlation a first-order risk not a detail of modelling. Suppose the book is levered by 2.45$\times$ based on $\bar{\rho} = 0.15$, and correlations move to 0.6. The realised risk then roughly doubles.

[Fact] Correlations rise in stress. Trend-following adds a second mechanism to the usual one. The strategy *selects for* correlated positions. A system holding 50 markets holds the ones that are trending, and markets trend together when a common macro factor moves. The effective N is smallest exactly when the shock arrives.

Monitoring, not forecasting. [Practice] **Recommendation: do not try to forecast correlation. Monitor three things continuously, and act on them:**

1. **Realised portfolio volatility against the target.** This is the direct measurement. If realised risk runs above the target, the correlation assumption is already wrong.
2. **The effective number of bets.** Take Σ, the covariance matrix of instrument returns, and the current position vector π_t. The portfolio variance $\pi_t^\top \Sigma \, \pi_t$ splits exactly across the eigenvectors of Σ, its principal components. Write ω_i for the fraction of that total contributed by the i-th component, so that $\omega_i \geq 0$ and $\sum_i \omega_i = 1$. Then $N_{\text{eff}} = 1/\sum_i \omega_i^2$ is the inverse Herfindahl index of those shares. It equals N when every component carries the same risk, and 1 when a single component carries all of it. It counts independent bets rather than tickers. A sharp fall in this number signals crowding inside the program's own book.
3. **Position concentration by sector and by macro factor**, such as duration, the dollar, equity beta and energy. The trend signal ignores these factors.

11.7 Crowding and liquidity

As §9.5 argued, crowding's first-order effect is on the loss distribution rather than the mean. The mechanism is simultaneous deleveraging. Many participants with similar positions and similar risk limits hit those limits at the same time, which turns a moderate adverse move into a large one.

Two defences do not require measuring crowding:

- **Stagger the program's own risk limits**, so that its deleveraging is not a step function. If the volatility target, the drawdown multiplier and the position caps all bind at once, the program has created its own version of the same problem.
- **Trade more slowly than other trend-followers.** The no-trade buffer of §8.7 has a useful side effect here. It keeps the program from competing for liquidity in the same minutes as everyone whose signal flipped at the same time.

One defence does measure crowding. **Maintain a generic trend replicator**, a simple, public, parameter-free version of the strategy, and monitor the correlation of the program's P&L with it. Rising correlation with the generic version means the program is converging on the consensus position. This gives warning before the unwind rather than during it.

11.8 The governance layer

Trend-following programs rarely end because of a modelling error. They end when someone decides to stop during a drawdown that was statistically unremarkable. The arithmetic of §5.9 makes the case. On the same iid-Gaussian baseline, a three-year window shows a negative realised Sharpe ratio with probability $\Phi(-S\sqrt{3})$, where S is the true annual Sharpe ratio and Φ the standard normal CDF. That is about one window in 24 at $S = 1.0$, and close to one in four at $S = 0.4$. Neither outcome is evidence of anything. Both last long enough to end a program.

[Practice] Decide the following in advance, in writing, before any money is at risk:

- **What drawdown is consistent with the strategy working.** Compute it from the simulated distribution, not from the historical maximum. The historical maximum is one draw, and it is almost certainly smaller than the drawdown that will eventually occur.
- **What evidence would constitute genuine deterioration** (§9.6), stated as a measurement rather than a P&L threshold. The variance-ratio monitor is the right instrument, because it does not depend on the program's own returns.
- **Who decides, on what schedule, with what information.** A review date set in advance prevents the worst decisions, which are the ones made on the worst days.

Governance belongs in a technical chapter. The strategy's statistical resolution is measured in decades, and its returns are right-skewed at the level of trades. So the governance rule *is* a risk parameter. It is one of the few risk parameters whose value the program controls.

§11 Key takeaways

1. The pitfalls at the research stage all reduce to one thing: information from the future, or from the drift, entering the test. Demean, and compute every estimate causally.
2. **Log every parameter change with a date.** A system re-tuned after each drawdown has an in-sample live track record.
3. Volatility targeting stabilises realised risk **[Fact]**, and it improves the Sharpe ratio where volatility predicts returns. It does not protect against volatility that jumps, and it may be a larger part of the premium than the signal is.
4. **Stops add expected return only under positive autocorrelation** (Kaminski and Lo). In a trend system they are a redundant fast signal. Keep a hard stop as an operational circuit-breaker, not as alpha.
5. Drawdown control turns temporary losses into permanent ones, and it interacts badly with volatility targeting. Use it gently, and justify it by business survival, not by return.
6. **The convexity is synthetic, and it fails at gaps.** Limit moves, weekend jumps and negative prices are the exposures. Decide in advance what the code does with each.
7. Correlation is a first-order risk, because the IDM sets leverage from the assumed correlation and trend-following *selects for* correlated positions. Monitor realised volatility and the effective number of bets, rather than forecasting correlation.
8. **Commit to the governance rule in advance.** At these Sharpe ratios, a multi-year losing run is unremarkable, and the decision to stop is the largest unhedged risk in the program.

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12. Synthesis

12.1 The framework in one page

Everything above reduces to three statements, each of them exact.

One. What it is. A trend-following system is a map from price history to position, increasing in recent returns:

$$\pi_t = c_t \, \frac{\sigma^\star}{\hat{\sigma}_t} \, g\Big(\sum_{k \geq 0} w_k \tilde{r}_{t-k} \Big)$$

Its four stages are the kernel, the response, risk scaling and aggregation. Every named indicator is a choice of the first two, and §7.2 gives the exact kernel for each.

Two. What it earns.

$$\mathbb{E}[\pi_t r_{t+1}] = \underbrace{\sum_{k \geq 0} w_k \gamma_{k+1}}_{\text{autocovariance}} + \underbrace{\mu^2 \sum_{k \geq 0} w_k}_{\text{drift}}$$

For the canonical rule, price minus its n-bar moving average on demeaned returns, the first term is exactly $\frac{1}{2}\gamma_0(\mathrm{VR}(n) - 1)$. **Trend-following is long the variance ratio.** Remove the drift term as a confound. Do not count it as a source of edge.

Three. What shape it is.

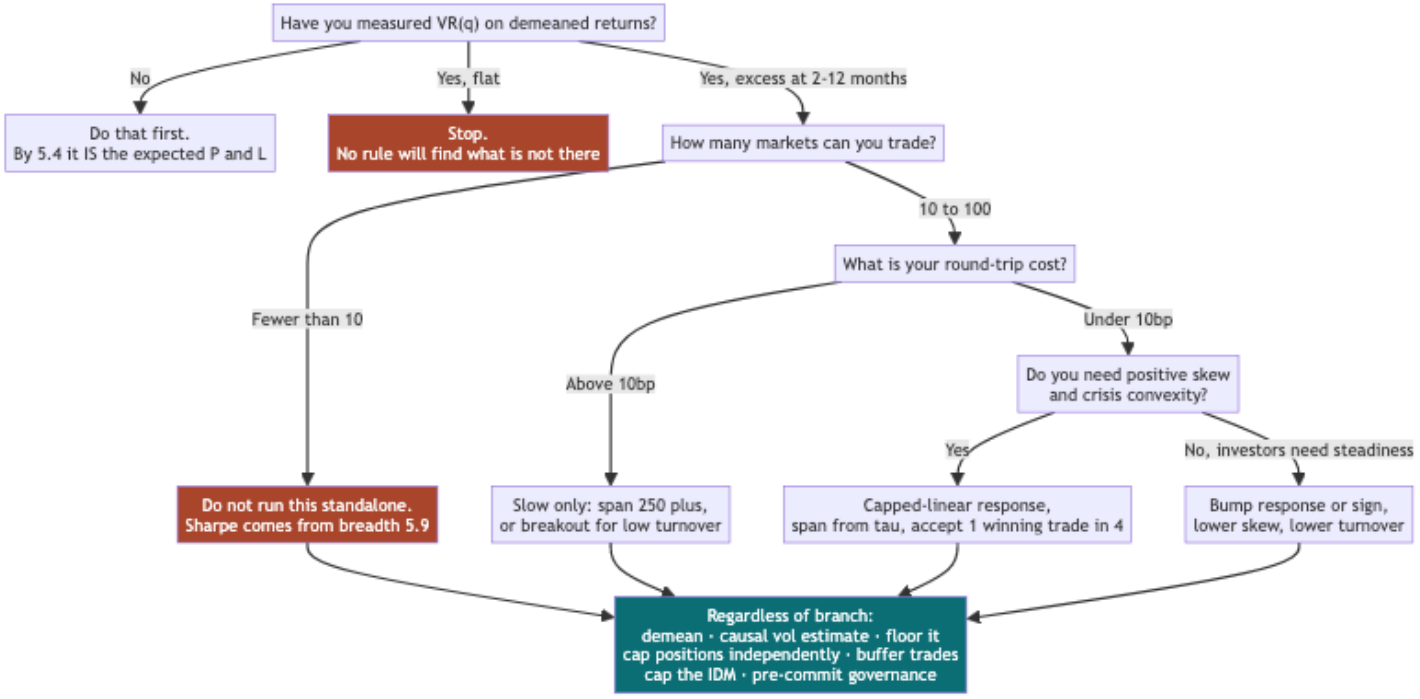
$$\sum_{t=0}^{T-1} x_t r_{t+1} = \underbrace{\frac{x_T^2 - x_0^2}{2\alpha(1-\alpha)}}_{\text{convexity}} + \underbrace{\frac{2-\alpha}{2(1-\alpha)} \sum_{t=0}^{T-1} x_t^2}_{\text{trend energy}} - \underbrace{\frac{\alpha}{2(1-\alpha)} \sum_{t=1}^{T} r_t^2}_{\text{premium paid}}$$

The strategy is **long trend energy and short realised variance**, in a ratio fixed by the speed of the filter. The convexity is real. The strategy pays for it continuously, and it fails at gaps because it is synthetic.

These three statements, plus the calibration of §5.9, are enough to analyse almost any question in the field. The calibration says that a diversified Sharpe ratio near 1.0 corresponds to a daily autocorrelation of about 0.0025.

12.2 Decision tree

The tree routes from the first measurement to a design, and ends in the steps that apply on every branch.



12.3 Building one from scratch

The build has stages, with a gate at each stage. Stages 1–3 are infrastructure. They are less interesting than the later stages, and skipping them is the usual cause of failure. Read the table as a *build* order. It does not show the ordering of effort of §8.1. A signal must exist before a portfolio can be built around it, which is why stage 4 comes before stage 5. Stage 4 should still take the least time.

Stage	Work	Gate before proceeding
1. Data	Futures panel, return-stitched, dead markets included, scheduled rolls	Return distribution shows no artefacts at roll dates; market count does not only rise
2. Measurement	VR(q) profile on demeaned returns, per market and pooled	A pooled excess exists at some horizon. If not, stop
3. Risk plumbing	Causal volatility estimates, position sizing, realised-risk monitor	On synthetic paths of known volatility, realised risk matches target within a few percent
4. Signal	One EWMA crossover, span from τ, capped-linear response	On synthetic paths, performance is 80–95% of the Kalman-optimal benchmark. Above that means look-ahead
5. Portfolio	IDM, sector caps, signed correlations	Realised portfolio volatility tracks target out of sample
6. Costs	Impact model, no-trade buffer, turnover budget	Sign survives 3$\times$ the cost estimate
7. Validation	Block bootstrap, null bootstrap, deflated Sharpe, walk-forward	Sharpe sits in the upper tail of the null distribution, not merely above zero
8. Risk	Gap stress tests, drawdown multiplier, crowding monitor, governance rule	Written, dated, and agreed before capital

12.4 Advice for someone starting today

1. **Measure the variance ratio before writing a backtest.** It is the expected P&L (§5.4). It takes an afternoon, and it shows whether to continue.
2. **Demean.** Every market, every test, always (§5.3). This one line removes the most common way that trend research misleads its author.

3. **Spend effort on the volatility estimate and the correlation structure**, not on the indicator. That is where the return comes from (§7.3).
4. **Set the lookback from trend persistence, not from backtest P&L.** Use a span of $\approx 2\tau$. The optimum is flat, so an approximate value is good enough. Optimising the lookback is a common cause of overfitting.
5. **Trade slowly.** Costs, not the quality of the signal, set the horizon (§8.8). Partial adjustment always beats full rebalancing (§8.7).
6. **Diversify across genuinely different markets.** Lowering $\bar{\rho}$ from 0.15 to 0.05 beats trebling the number of markets.
7. **Expect to be wrong three times out of four.** The hit rate is a design choice and not a measure of quality. The strategy's entire profit comes from a handful of trades (§5.7).
8. **Test on synthetic data where the answer is known.** Beating the theoretical optimum proves a bug, and it is the best detector of look-ahead available.
9. **Cap positions independently of the volatility scaling.** A collapsed $\hat{\sigma}$ should produce a capped position, not an unbounded one.
10. **Write the governance rule before any money is at risk.** At this Sharpe ratio, a three-year losing run is unremarkable, and the decision to stop during one is the largest risk in the program.

12.5 What is and is not known

Established. The algebra is settled. The P&L identities of §5.2, §5.4 and §5.5 are theorems, and no one disputes them. The kernel equivalences of §7.2 are exact. The statistical calibration of §5.9 follows from arithmetic. The effect is undetectable market by market and visible only in aggregate. Replicated evidence shows that volatility scaling stabilises realised risk, and that it improves risk-adjusted performance where volatility predicts returns (§11.2). The convexity of the payoff in the underlying's move can be derived, and so can the dependence of the hit rate on the response function. Both have been derived.

Contested, with arguments on both sides that deserve respect.

- Whether the time-series momentum premium survives the drift correction (Huang and co-authors against Moskowitz and co-authors).
- How much of the premium is really the volatility scaling (Kim, Tse and Wald).
- Whether the premium has decayed since 2009.
- What CTA investors have actually earned net of fees and biases. Here the critique (Bhardwaj, Gorton and Rouwenhorst) is stronger than the defence.

Not known. Why trends exist is unknown. Five families of mechanism have support, and none is decisive. None of them comfortably explains the cross-section of results, in which trend-following works well in the most information-efficient markets in the world. No reliable method exists to detect the exhaustion of a trend, to measure crowding in real time, or to forecast when the strategy will work.

The largest open risk in the field, in the view taken here, is methodological. Essentially all of the long-history evidence that trend-following persists comes from firms that sell it. This is not a reason to dismiss the evidence, because these firms are the only groups with the data. Two of them, with different books, reached the same answer independently. But this conflict means that the most valuable contribution to this literature would be a long-history study, on demeaned returns, by an author with no position.

The view taken here is that trend-following remains the most intellectually honest strategy in systematic trading. Its mechanism is a measurable statistic, its payoff shape is an algebraic identity, and its failure modes are known and can be enumerated. Whether it still pays enough to be worth running is a separate question. The variance-ratio monitor of §9.6 will answer that question long before the P&L does.

§12 Key takeaways

1. Three exact statements describe the field. A trend system is a map from price history to position. Its expected P&L is long autocovariance and, for the canonical rule, exactly half the excess of the variance ratio over one. Its realised P&L is long trend energy and short realised variance.
2. Measure the variance ratio and demean before any backtest. Then spend effort on volatility scaling and the correlation structure, which is where the return is.
3. The algebra is settled. Why trends exist, and whether the premium has decayed, are not. Write the governance rule before capital is at risk.

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